INFO30008 Interactive Technology Project

A2 Group Report

The Design Behind *Cove:* Supporting for Task Initiation, Focus, and Healthier Scrolling Habits

Group 14

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Ideation process

FigJam ideas → Interview findings → Feature priorities → Early wireframes → Peer feedback → Final prototype

We started our ideation process by going back through our interview notes and the research we had collected. One thing that came up quite often was that students usually know when they should be studying, but still end up opening social media when they feel bored, stressed or do not know how to start a task, stating "*Putting my phone away worked for a little while but I would just get it back and continue doom scrolling. I need something that blocks my app usage completely and doesn't just let me bypass the blocked screen*" (Al-Mahdi et al., 2026). This has influenced our idea of not wanting the app to only block people from using their phones, but how it would help users notice their habits and make it a little easier to get back to what they were doing.

The first step was placing each of our different ideas into FigJam. Some of these were app limits, task lists, smaller subtasks, focus sessions with friends, reflection questions and a pet that could encourage the user. We also came up with more playful ideas, like earning coins, buying items from a shop, decorating a room and changing the pet's appearance. The diverse ideas that we came up with gave us a useful starting point and helped us see the different directions the app could take.

We then talked about which ideas actually connected to our research and which ones were realistic for us to prototype. We decided that tasks, Pair Up, reflection and the lockout feature were the most important. The shop, furniture and more detailed pet features became lower priorities. We liked these ideas, but the app was starting to feel too focused on the game elements. Since it is still meant to be a productivity app, we kept the pet and coins as smaller forms of encouragement instead of making them the main focus.

After this, we divided the features between group members. Each person made user flows and wireframes for their section, and we added them to the same Figma file. This made it easier to compare our work, although it also showed that some screens did not match at first. We had different colours, navigation layouts, button styles and ways of writing instructions. We gave each other comments and gradually made the screens more consistent. We ended up describing the overall style as "cute but disciplined". The pet gives the app some personality, while the layout still needs to feel clear and useful.

The task section also changed a few times. The early version was quite simple and mainly showed a list of tasks. Later, we added categories, due dates, estimated focus time and optional subtasks. We also considered what would happen when a task became overdue, so we added options to continue or reschedule it. Some users thought that creating subtasks was required, even though we intended it to be

optional. We therefore made the wording clearer and reduced some of the steps. We also added small coin rewards for completing subtasks.

Once the sections were connected into one prototype, we tested the main flows with users. Their feedback showed that the general idea was understandable, but some parts felt restrictive, especially the reflection and focus-session rules. We reviewed these areas and tried to give users clearer choices. The final design still keeps some friction to interrupt impulsive scrolling, but it also gives users more control over what happens next.

Main Design Concept

Cove replaces restriction with honest reflection. It interrupts digital avoidance at the exact moment it happens, responds in a way matched to the situation, and asks for real evidence of engagement, not just time spent, before any reward is given.

The paradox for this idea would be that students want to focus, yet the devices they study on are built to pull them away. Hard blockers are easily bypassed and breed resentment, while gamified study apps often reward time spent rather than real work. Support must be firm enough to interrupt impulse, yet gentle enough that users do not disregard. Cove resolves this by keeping the user in control throughout where restriction is replaced with reflection, so the intervention holds without becoming something to escape.

There are two core ideas to Cove's direction. The first one is a reflective scroll interruption that tackles avoidance at the moment of distraction. During onboarding, users choose which apps to restrict and set time limits. When a restricted app is opened, even with time remaining, Cove responds to the situation rather than applying one fixed rule. It asks why the user is there. Someone taking a break completes a short self-affirmation task before entering, turning passive scrolling into a deliberate choice. Someone admitting avoidance receives tips on getting started, or is redirected to a task on their to-do list. If a task is due soon, Cove names it directly and if the user has previously regretted using the app, Cove invites them to reconsider. Afterwards, it asks whether the session was needed, helping users recognise their own patterns over time. Because each response adapts to the scenario, the intervention stays relevant rather than becoming a prompt users learn to ignore.

The second core idea is supported task initiation. Starting is often harder than continuing. Users add tasks to a to-do list and begin a focus session individually, in private rooms with friends, or in public rooms alongside strangers. Shared presence creates gentle accountability without comparison. Users who feel

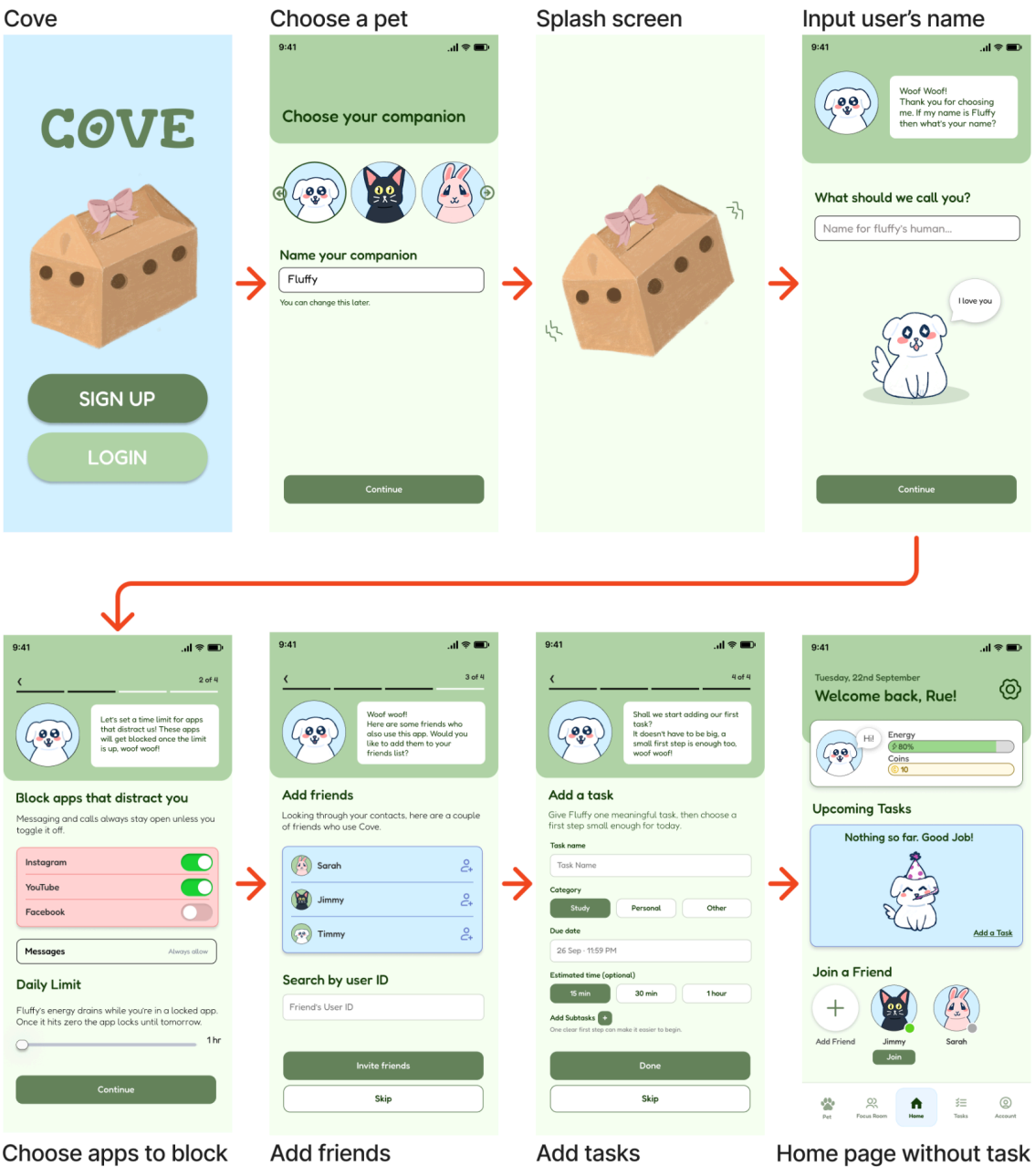
stuck can access tips on starting or continuing, and if they leave a session abruptly, Cove asks why in a way that offers support for avoidance and no penalty for genuine emergencies. Completing a session requires a short 50–100 word summary of what was learned, so rewards reflect real engagement rather than elapsed time.

The ideas are supported by the cosmetic of a pet that acts as a non-punitive guide, grown through points from genuine work. This allows users to have genuine incentive to use the app long term when productivity gets difficult.

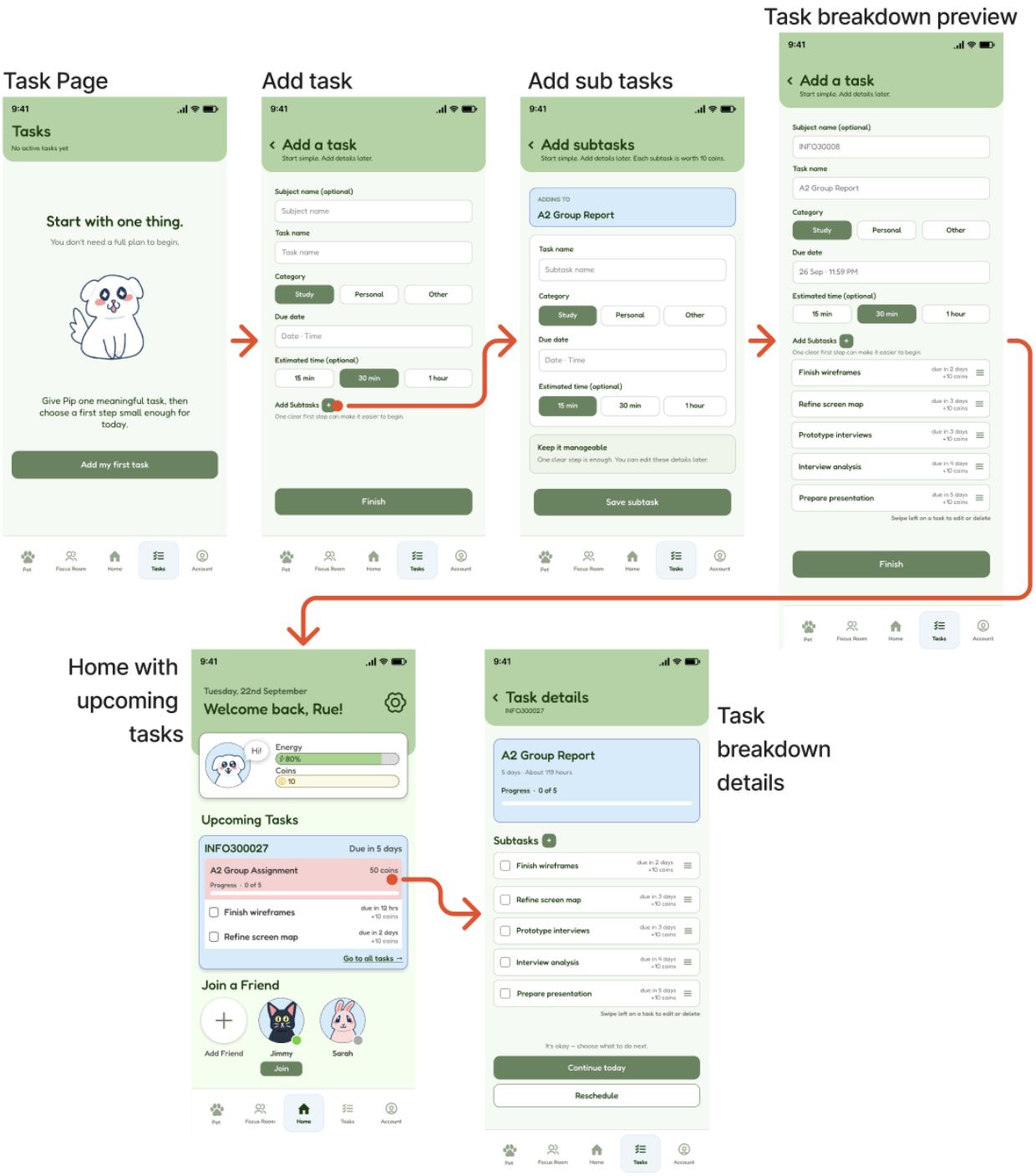
Cove draws on Sirois and Pychyl's (2013) view of procrastination as mood regulation and the JITAI framework (Nahum-Shani et al., 2018). Its "taking a break" pathway requires typed self-affirmation, following Xu et al. 's (2024) TypeOut, to turn reflexive phone use into deliberate choice. The "avoidance" pathway offers tips or redirects, addressing stalled work intentions. Our findings show restriction fails without genuine reflection.

Detailed Prototypes

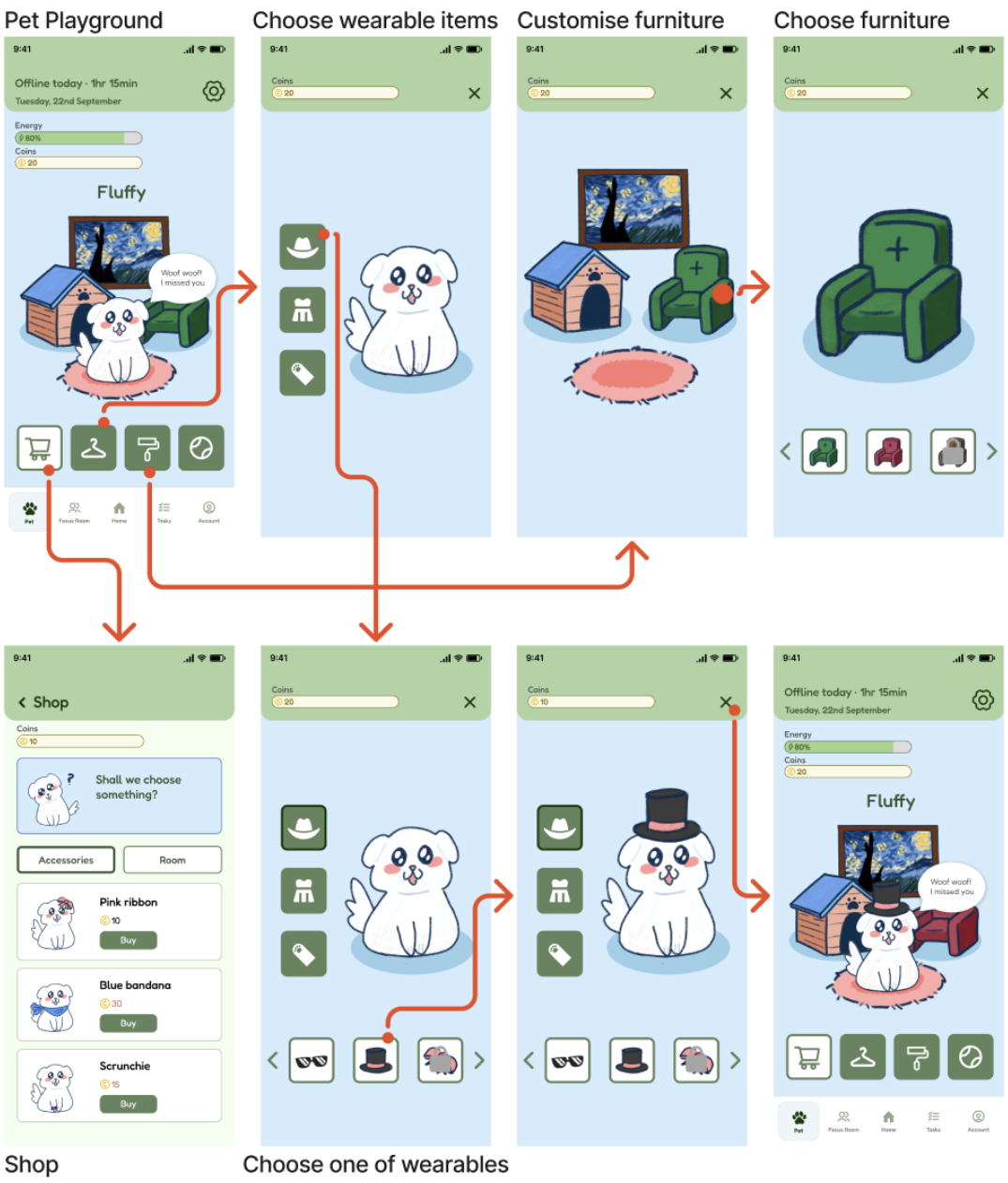
Onboarding



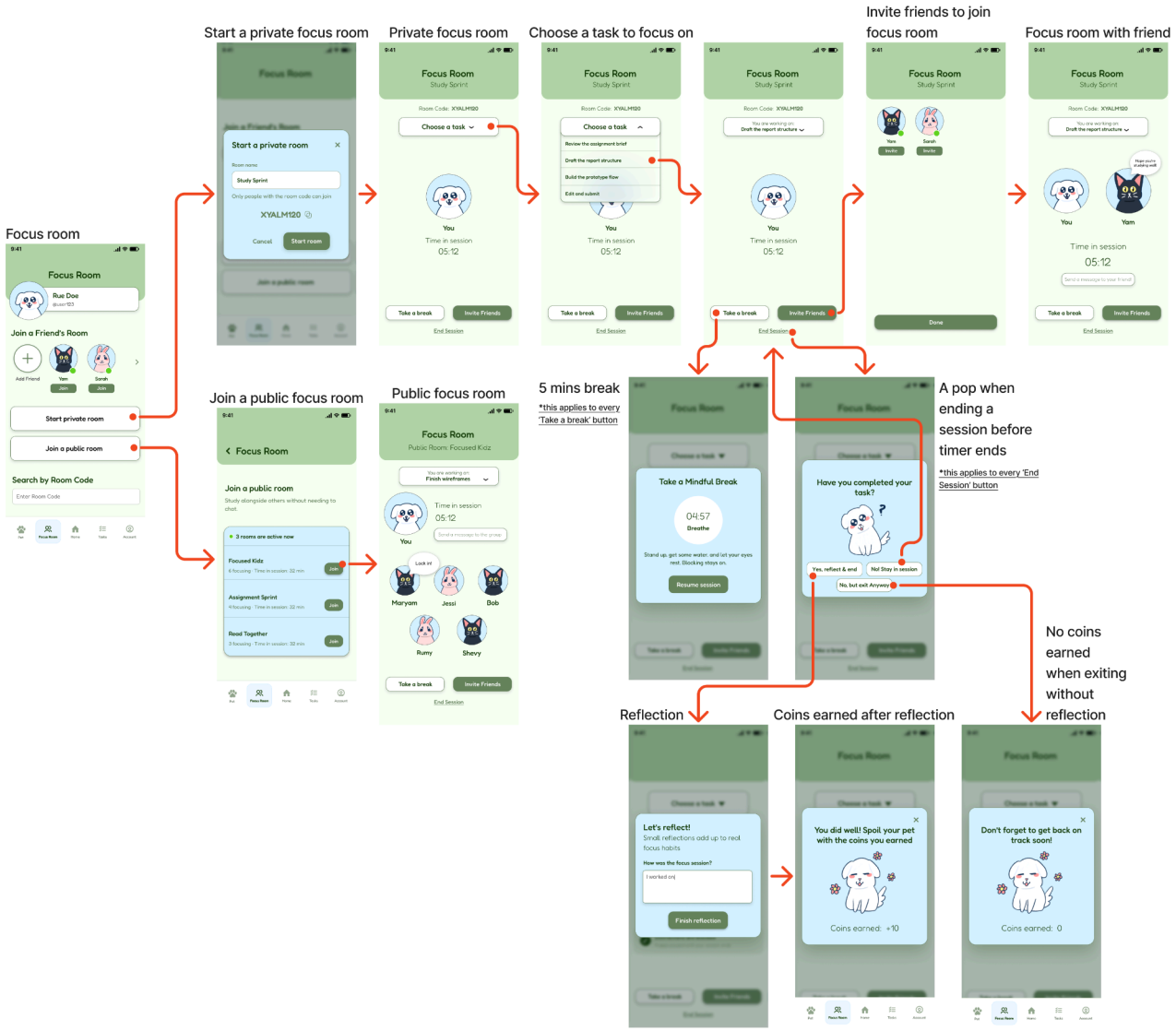
Tasks


Task Page
Add task
Add sub tasks
Task breakdown preview
Home with upcoming tasks
Task breakdown details
Tasks
Start with one thing.
You don't need a full plan to begin.
Give Pip one meaningful task, then choose a first step small enough for today.
Add my first task
Add a task
Subject name (optional)
Task name
Category
Study
Personal
Other
Due date
Estimated time (optional)
15 min
30 min
1 hour
Add Subtasks
Finish
Add subtasks
A2 Group Report
Keep it manageable
Save subtask
INFO30008
26 Sep · 11:59 PM
Finish wireframes
Refine screen map
Prototype interviews
Interview analysis
Prepare presentation
Finish
Tuesday, 22nd September
Welcome back, Rue!
Energy
Coins
Upcoming Tasks
INFO300027
Due in 5 days
A2 Group Assignment
50 coins
Finish wireframes
Refine screen map
Go to all tasks →
Join a Friend
Add Friend
Jimmy
Sarah
Join
Task details
A2 Group Report
Progress · 0 of 5
Subtasks
Continue today
Reschedule

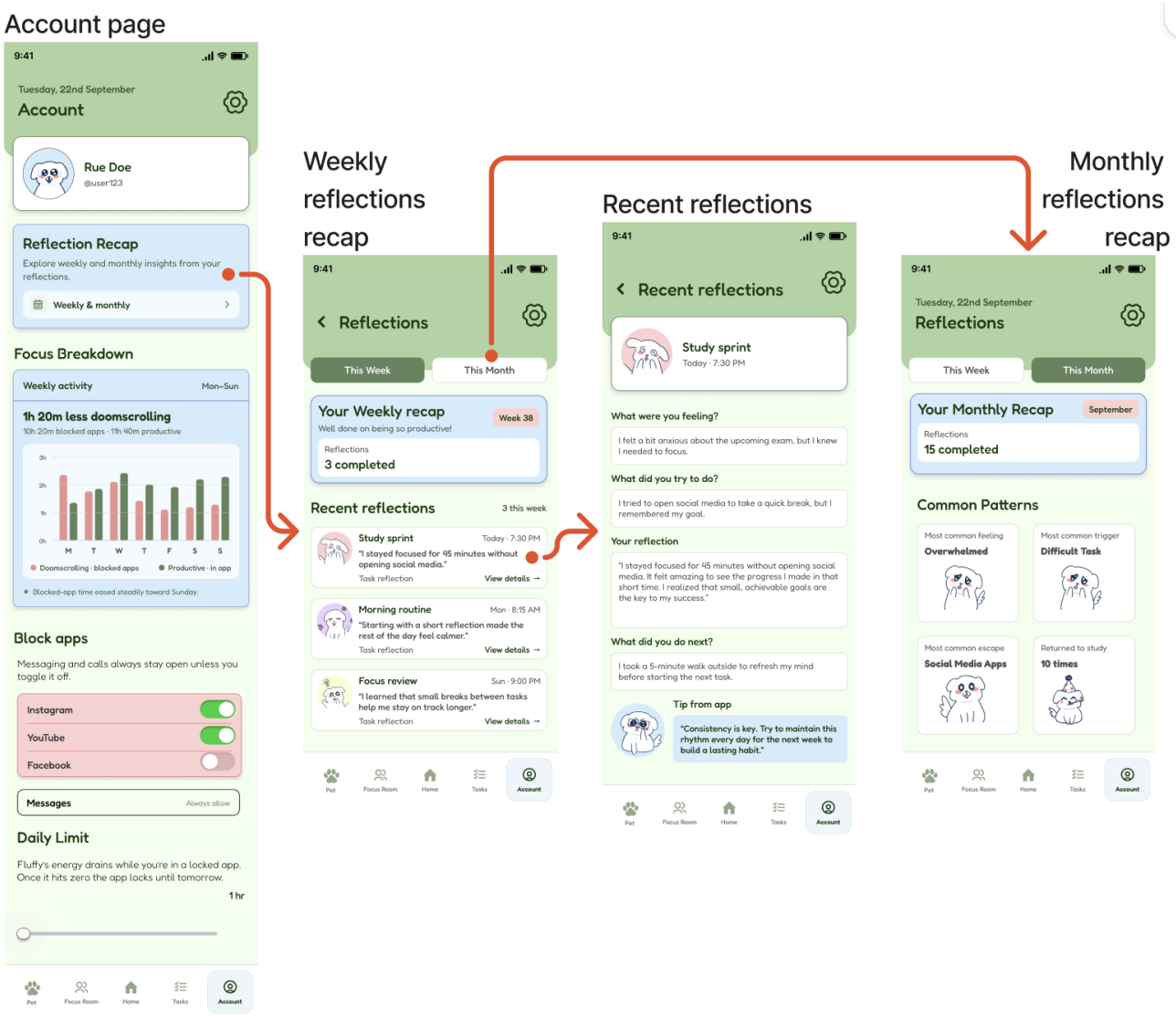
Pet Playground



Focus Room

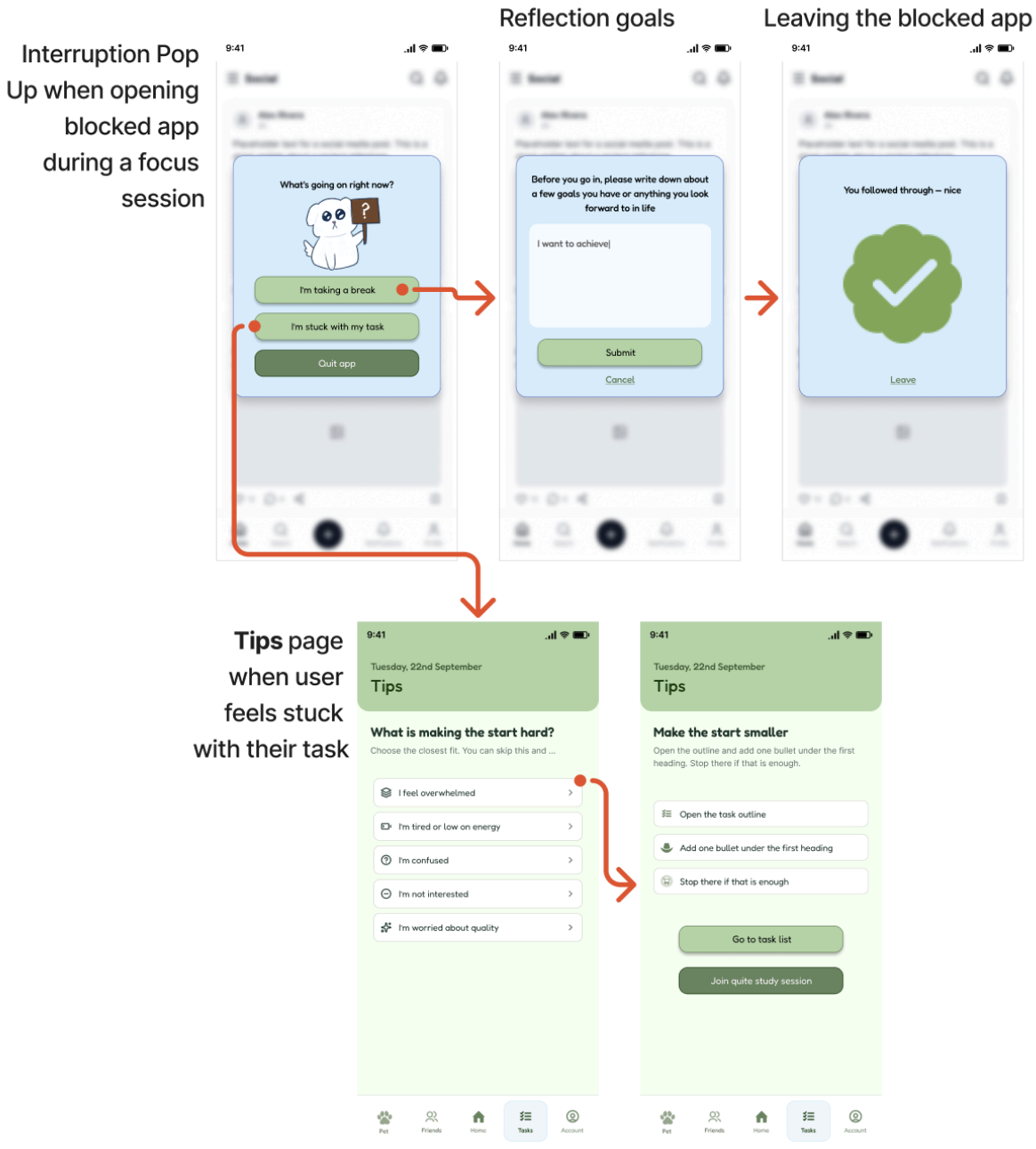

Focus room
Start a private focus room
Private focus room
Choose a task to focus on
Invite friends to join focus room
Focus room with friend
Join a public focus room
Public focus room
5 mins break
*this applies to every 'Take a break' button
A pop when ending a session before timer ends
*this applies to every 'End Session' button
No coins earned when exiting without reflection
Reflection
Coins earned after reflection

Account Page

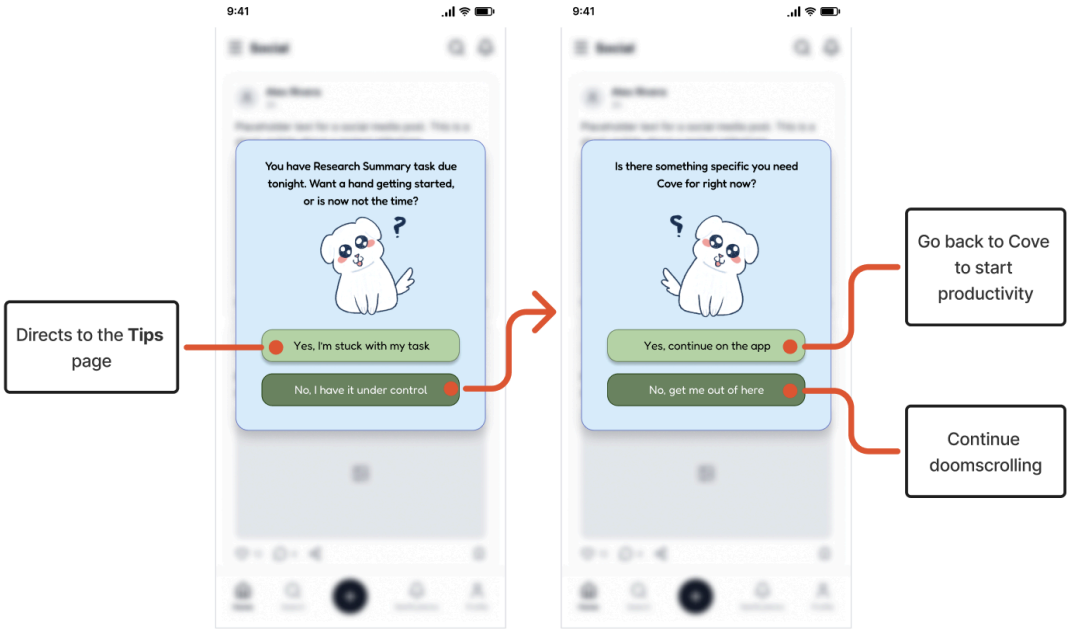


Locked-Out Pop Ups

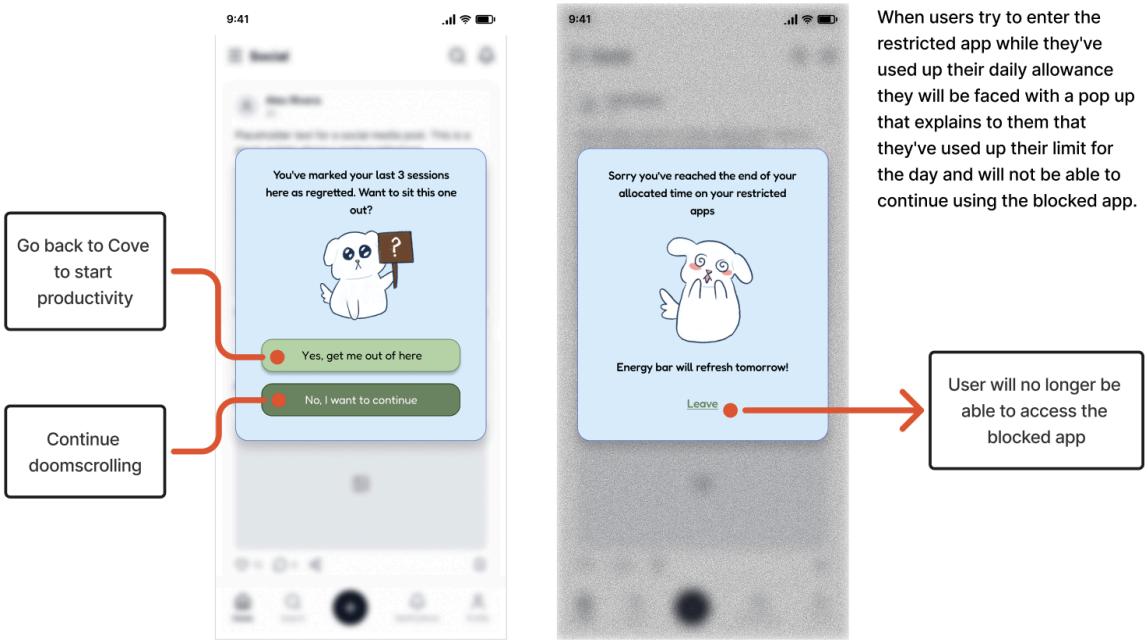
Locked-Out #1: General circumstance of opening a blocked app while there is time allowance left.



Locked-Out #2: In the case that a task is due within 24 hours.

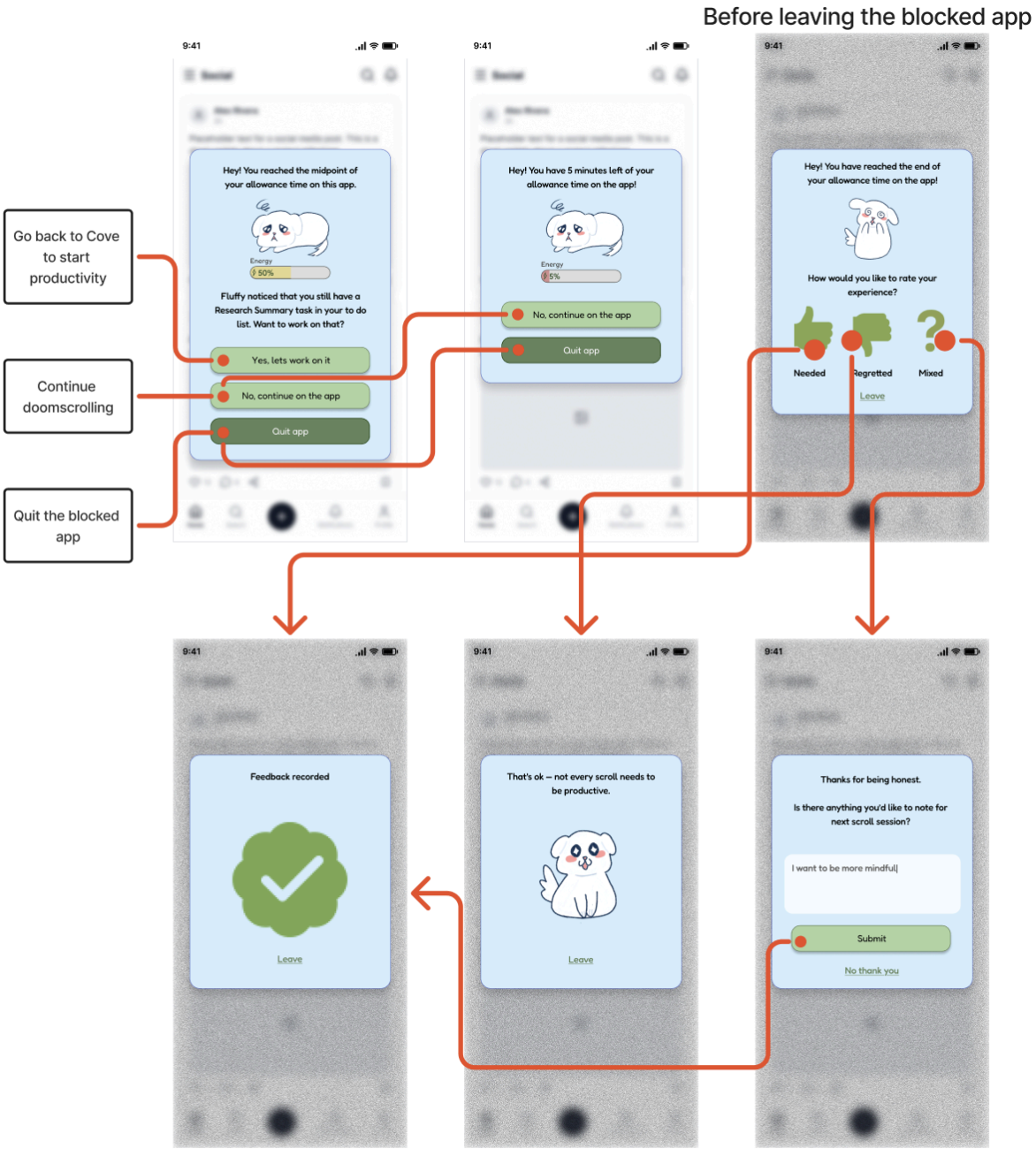


Locked-Out #3: In the case that a user has regretted using a blocked app repeatedly in the past.



When users try to enter the restricted app while they've used up their daily allowance they will be faced with a pop up that explains to them that they've used up their limit for the day and will not be able to continue using the blocked app.

Locked-Out #4: Notifications when in app while the time allowance on blocked app runs out.



Link to Figma prototype:
https://www.figma.com/proto/Vkt3tuczadGLkEtaFEn1zK/Cove?node-id=2668-13421&p=f&t=i43eNk3nPxSt9FaZ-1&scaling=scale-down&content-scaling=fixed&page-id=2668%3A5319&starting-point-node-id=2668%3A13421&show-proto-sidebar=1

User-Based Evaluation Plan

This evaluation tested whether Cove's main design concept works for real users. We focused on how well participants completed five task scenarios, covering onboarding and Cove's four core features. It aimed to answer four questions:

- Can users complete each scenario without help?
- Do users understand how the features connect?
- Does writing a reflection before a reward feel fair rather than like a chore?
- Do the pet and gentle nudges motivate users, or feel annoying?

Participants

Each of the seven group members tested 1-2 participants, 11 in total. All were friends of group members aged 18-24: 9 were university students, 2 were final-year high school students and. All said they often use their phone to avoid tasks.

Task Scenarios (table)

Feature	**Task scenario**	**What it tests**
Onboarding	A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	Can users complete setup and understand the app restrictions and time allowance?
Tasks	You have an upcoming assignment and want to break it down into manageable steps so you can stay focused and avoid procrastinating.	Can users create a task, add details and understand how subtasks work?
Starting a Focus session	You have an assignment to work on and want to avoid	Can users start a session and understand the private and

	distractions while studying. Start a focus session and choose the setup that best suits your study preference.	shared room options?
Pet Playground	You have completed a focus session and earned coins. Explore the Pet Playground to see how you can use your rewards to interact with or customise your pet.	Can users find the Playground and understand how to spend their coins?
Locked- intervention	You are in the middle of a focus session when you feel the urge to leave Cove and open a distracting app. Follow the prompts and see how Cove responds to your attempt to leave the session.	Do users understand the prompts and what happens when they continue or leave?

Procedure

Group members ran the sessions either on their own or in pairs, using the Figma prototype. Participants worked through the five scenarios in the app's natural order. We noted where they hesitated, took an unexpected route or misunderstood a label. Their comments helped us distinguish between difficulty using a feature and simply not liking how it worked. Each scenario was followed by a difficulty rating from 1 (very easy) to 5 (very difficult). All 11 participants also completed the SUS questionnaire after testing. Depending on the session, we kept written notes, audio recordings or screen recordings. We then added our observations, ratings and participant comments to the shared template so the group could compare findings across sessions.

Data collected and how

All members used a shared template for each scenario: observed behaviour (hesitations, errors, unexpected paths), participant feedback and quotes, a usability insight and the difficulty rating, in addition to each participant's background and SUS score. Data was recorded through a mixture of audio, written notes and screen recording.

What insights the data gives

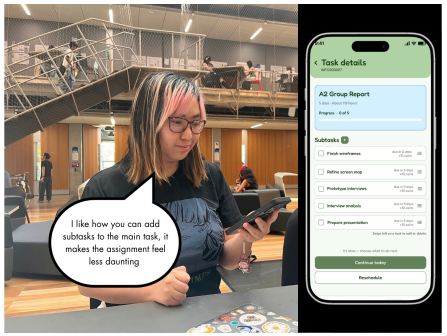
Difficulty ratings taken straight after each task show which scenarios users found harder. SUS scores give an overall measure of usability, and we compared Cove's average with the benchmark of 68, drawn from 446 studies (Sauro 2011).

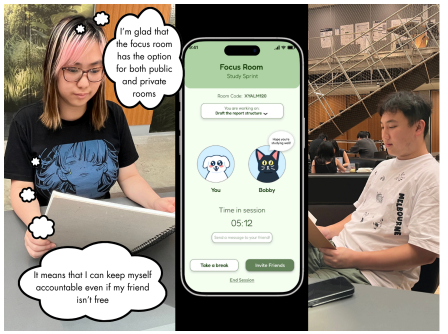
The numbers show where problems are, and observations and quotes explain why. We grouped these through affinity mapping, counting how many participants raised each issue, so themes reflect patterns rather than one-off comments. This reveals usability issues and how users respond to Cove's core idea: whether the reflection step and nudges help them pause before giving in, or feel frustrating enough to skip.

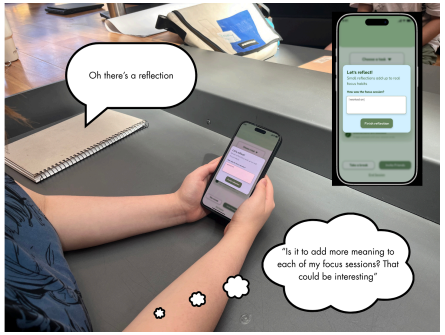
Ethics and limitations

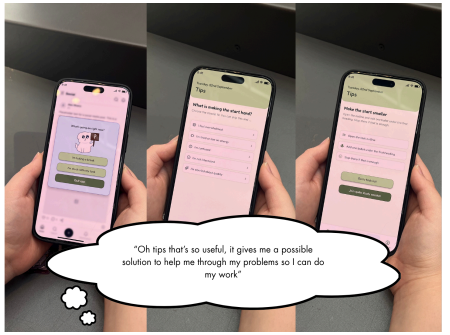
Testing with friends likely inflated ratings: Dell et al. (2012) found participants were about 2.5 times more likely to prefer a design they believed was the researcher's. We therefore gave observed behaviour more weight than ratings. Facilitators sometimes prompted participants, and scenarios always ran in the same order, so later scenarios may have benefited from learning. Participants consented using the group's Consent Form and Plain Language Statement, are anonymised, and data is stored in the group's protected FigJam.

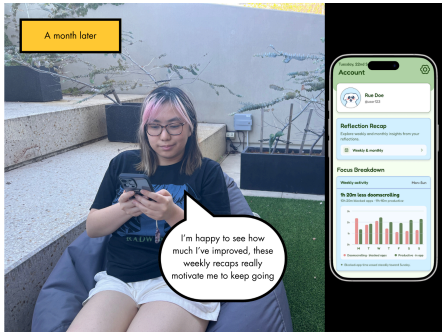
User Journey











References & Statement of AI use

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Nahum-Shani, I., Smith, S. N., Spring, B. J., Collins, L. M., Witkiewitz, K., Tewari, A., & Murphy, S. A. (2018). Just-in-time adaptive interventions (JITAIs) in mobile health: Key components and design principles for ongoing health behavior support. *Annals of Behavioral Medicine, 52*(6), 446–462. https://doi.org/10.1007/s12160-016-9830-8

Sauro, J. (2011). *A practical guide to the System Usability Scale: background, benchmarks & best practices*, Measuring Usability LLC, Denver.

Sirois, F., & Pychyl, T. (2013b). Procrastination and the priority of short-term mood regulation: Consequences for future self. *Social and Personality Psychology Compass*, *7*(2), 115–127.

Xu, X., Zou, T., Xiao, H., Li, Y., Wang, R., Yuan, T., Wang, Y., Shi, Y., Mankoff, J., & Dey, A. K. (2022). TypeOut: Leveraging just-in-time self-affirmation for smartphone overuse reduction. *CHI Conference on Human Factors in Computing Systems*, 1–17. https://doi.org/10.1145/3491102.3517476

Individual Components

Bowen Mai

Task Scenario

Alex — The Uncertain Starter

Alex is a 20-year-old university student who wants to make progress on her assignments but often gets stuck deciding where to begin. She usually understands what she needs to submit. An open-ended essay question, however, can leave her unsure which reading to use or what her first paragraph should say. She rereads the instructions and thinks through different possibilities while the document stays blank. During this hesitation, checking her phone becomes an easy way to put the decision off.

Alex finds it easier to begin when she has a small goal with a clear endpoint, such as reading one introduction and making three notes. Once she starts, unfamiliar ideas can still interrupt her progress. A confusing paragraph may make her question her choice of article and reach for her phone again. She needs help recognising what has stopped her and finding something she can do next. Making a rough note or saving a question for her tutorial allows her to keep working without resolving every difficulty immediately.

Alex wants a study tool that makes the next action easier to choose. She appreciates having the option to study alone or with company, depending on what she needs that day. She prefers brief check-ins and simple task planning because too many decisions can delay the work further. Pet rewards are enjoyable, but she also wants to see progress in the assignment itself. A useful session leaves her with a few notes she can build on and a clearer idea of what to do next.

Task scenario

One afternoon, Alex sits down in the university library to work on an essay. Her laptop is open, and she has the assignment instructions beside a blank document. She reads the question twice but still cannot decide what her opening paragraph should say. She starts checking her phone while telling herself that she will begin in a minute. After a few minutes, she opens Cove. She has already selected the apps that usually distract her. For this session, she wants to make some notes about one reading so that she has something concrete to use when she starts drafting.

Alex adds the essay to her task list. Looking at the whole assignment still feels uncomfortable, so she opens the task details and adds a smaller subtask: read the introduction of one article and write down three useful points. She leaves the rest of the assignment planning for later. This gives her a short piece of work she can recognise and attempt. She does not need to decide the structure of the entire essay before opening the reading. Once the subtask is saved, she returns to the task list and chooses it as the starting point for her study session.

She opens a private Focus Room and starts working on the reading. Today she would rather work quietly than join other people, although a shared room is available when she wants company. With her task chosen, she puts the phone beside her laptop and begins reading the first page. She writes one rough note, then reaches a paragraph that she cannot understand. She reads it again and starts wondering whether she has chosen the wrong article. The familiar urge to escape returns. Without deciding to take a proper break, she picks up her phone and opens an app she usually scrolls through.

The blocked-app intervention interrupts her attempt to keep scrolling. Alex returns to Cove and looks at the tips for getting back to her task. The interruption gives her a moment to notice what happened: she reached a difficult paragraph and moved away from it. She does not need another reminder of how much work remains. She chooses a smaller action she can take now, going back to the article to underline the sentence she does understand. She can leave a question beside the confusing part and bring it to her next tutorial instead of trying to solve everything immediately.

Alex returns to her notes and writes the remaining two points in her own words. They are not ready to go into the essay, but she can use them to begin a paragraph later. When the focus session ends, she

completes a brief reflection. She notes that she started scrolling when the reading became confusing and that writing a rough note helped her return. She keeps the response short because she does not want another writing task after working on the essay. Before moving on, she looks over her notes and marks the sentence she wants to use in her draft.

She then checks the coins she has earned and visits the Pet Playground. Alex looks at the available items and considers using her coins to customise her pet. She enjoys looking around for a moment, although the essay is still on her mind. After leaving the Playground, she returns to her task list and adds her next small step. The difficult paragraph in the reading is still marked with a question for her tutorial. She opens her essay document again, copies across her first note, and uses it to write a rough topic sentence for the essay's first paragraph.

Design precedent

One sec is an app that adds a pause when someone opens a selected app, such as social media. A breathing prompt appears before the user decides whether to continue. The reference image shows this prompt. One sec also supports intention tracking and app blocking. Its approach is relevant to Cove because it interrupts the familiar action of reaching for an app without thinking about why.

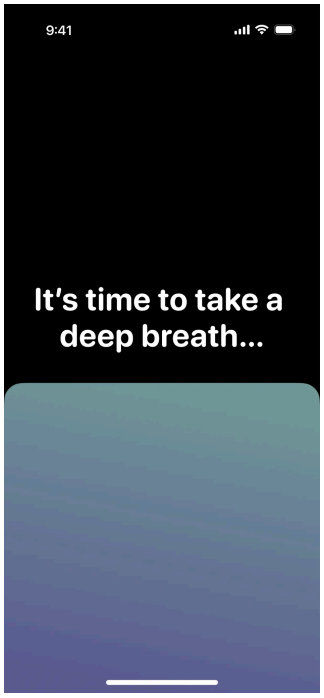
The principle behind this is purposeful friction. A small delay gives the user time to reconsider an impulsive action. Grüning et al. (2023) tested a version combining a delay, a deliberation message and an option to dismiss app opening. Their controlled experiment found that dismissal had the strongest effect; delay also helped, while the message alone was ineffective. The result suggests that an interruption needs a clear choice the user can act on.

My earlier Intention Checkpoint concept used a similar decision point. Students stated a purpose or made an initial attempt before accessing rewarding content. Their intention stayed visible, and the system guided them back to the original task. One sec helps explain why this pause is useful to examine. It gives students a chance to reconsider while preserving useful access to information and social connection, without suggesting that every phone interaction is a personal failure.

Cove can connect the pause to what the student wanted to do. A student who feels stuck could choose a smaller step, receive reassurance or take an intentional break. The companion can make the interaction feel supportive, while task breakdown and Focus Rooms offer ways back to study. This develops my earlier idea by pairing awareness of an intention with help acting on it.

However, another interruption can become frustrating. My earlier evaluation raised concerns about compulsory writing when someone has no initial idea. Cove's group feedback also suggested that long, repeated reflection could be frustrating. A brief checkpoint, an optional response and a clear next action would address these concerns. These remain proposed refinements based on feedback, and their effect on behaviour still needs testing.

Research on one sec supports examining this approach, but it does not establish Cove's effectiveness. For Cove, the useful questions are whether students understand their choices, return to an intended task and continue to find the interruption acceptable after repeated use. These outcomes would show more than whether a student can get through the prompt once.



one sec precedent. The breathing prompt creates a pause before the user continues. Source: one sec, https://one-sec.app/

Usability rational

Affordance

I would judge Cove's usability by whether a student can choose a manageable task, start focusing, see their progress and recover from a distraction. Affordance is useful here, although an action being available does not guarantee that people will notice it. Norman explains that actual and perceived affordances can differ, especially when an interface relies on conventions that users have learned elsewhere (Norman, 2008a).

A focus control needs a clear label and appearance so students know what will happen when they tap it. Norman calls these cues signifiers (Norman, 2008b). I would keep action labels and button styles consistent. The pet needs the same care. Its illustration might suggest companionship, but students still need to know whether tapping it opens customisation or progress. If part of the pet is interactive, that should be clear; decoration should not look like a button.

Breaking a task into smaller steps can help students keep track of what comes next. I would show the selected task near the control that starts a session and use the same terms across solo sessions and Focus Rooms. This follows Nielsen's principles of recognition, consistency and visible system status (Nielsen, 1994). A follow-up test should check whether students understand the revised controls as intended.

Feedback should tell students what their actions changed. They need to know whether a session is running, paused, interrupted or finished. A short message could explain why coins were earned. The pet's reaction can make that feedback feel friendly, but the important details should also be clear without depending on animation or colour. During follow-up testing, I would ask students to explain the current state and what they can do next without needing help.

Blocking deliberately slows users down, which makes control over the interaction especially important. My earlier Intention Checkpoint introduced a pause before returning to distracting activity, but interview

feedback questioned compulsory writing. I would keep reflection short and optional, with a clear way to continue without writing. Students also need an understandable exit when something interrupts their session. These choices follow Nielsen's principle of user control and freedom (Nielsen, 1994).

I would test these ideas by asking students to start work, join a room, explain a reward and respond to an interruption. Their hesitations, mistaken taps and explanations would help me assess whether the signifiers work. The group testing provides useful feedback; these proposed changes still need a follow-up usability test.

Wire-frames

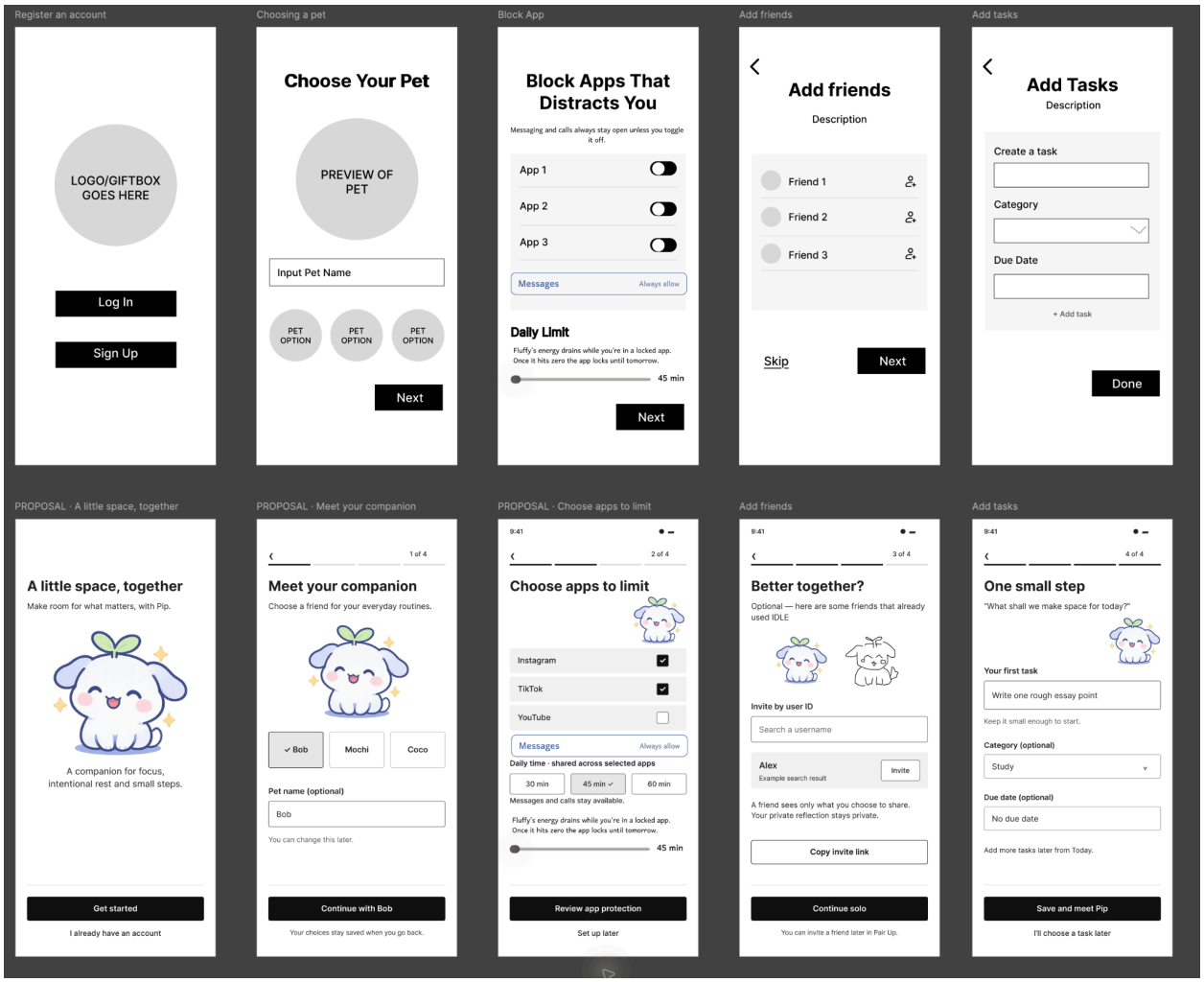


Figure 1. Choosing a companion. The outline shows which pet is selected. The preview and name field let the user check their choice before continuing. The name can be changed later. Source: team Cove prototype, Pet choosing.

Figure 2. Playground. Fluffy sits in the room, with energy and coins shown above. The four icons open the shop, dress-up, room customisation and play. Source: Figma final designs, playground.

Prototype screens for customisation

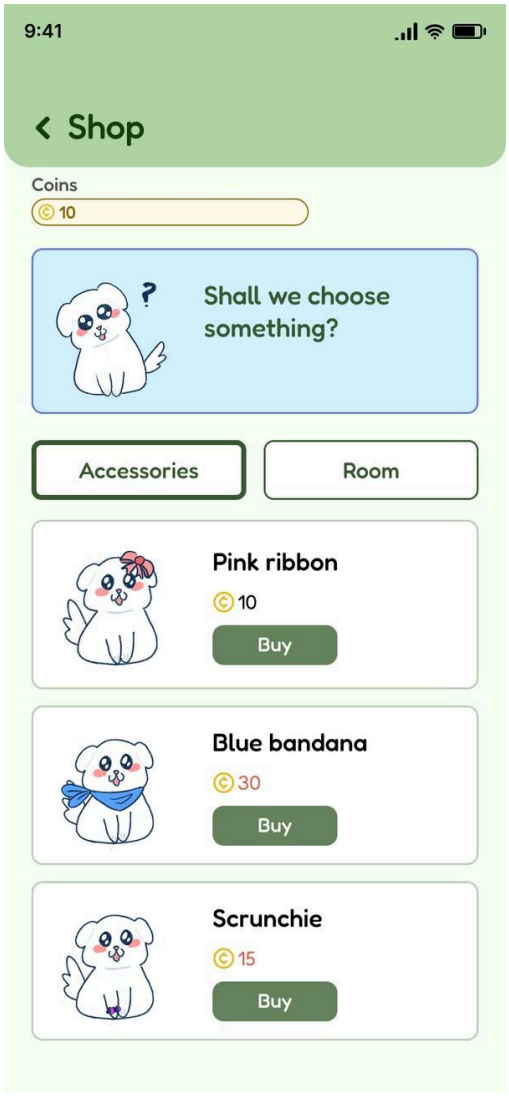
The dress-up and room screens keep the existing pet and furniture artwork. I would check whether students can find the controls, recognise a selection and leave the screen.

Figure 3. Dress-up. Fluffy stays in the middle of the screen, with three category icons for customisation. The coin balance and close button remain visible. Source: Figma final designs, dress up.

Figure 4. Room customisation. The house, picture, chair and carpet appear as separate objects. I would check whether students recognise them and can tell which one is selected. Source: Figma final designs, room cus 1.

Prototype screens for rewards and reflection

The shop shows what coins can buy. The end-of-time prompt asks users to reflect briefly on their scrolling. These screens show the design, while its effect on behaviour still needs testing.



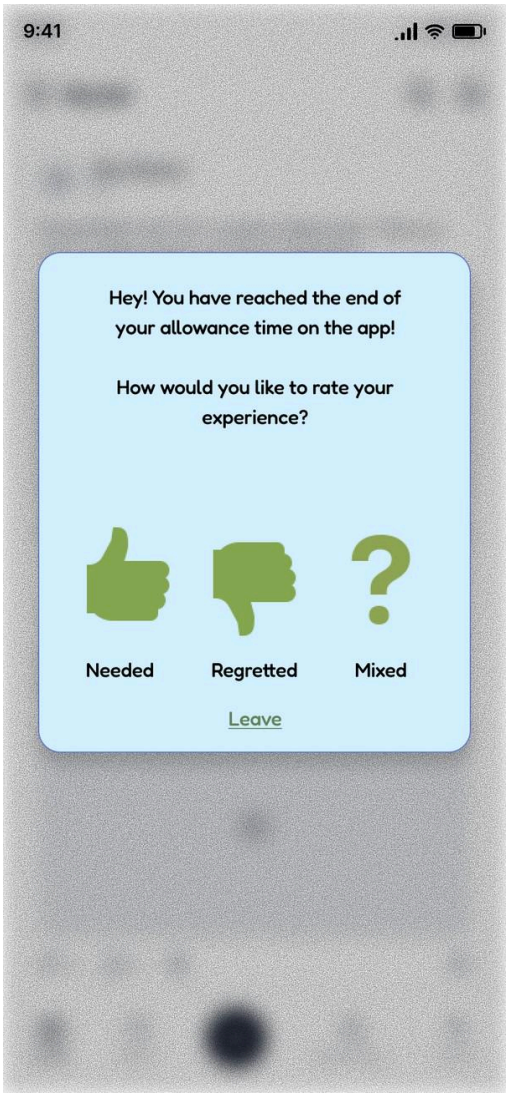


Figure 5. Accessory shop. The coin balance sits above the Accessories and Room tabs. Each item has an image, a price and a Buy button, so users can see what their coins can buy. Source: team Cove prototype, shop.

Figure 6. End-of-time prompt. Users can choose Needed, Regretted or Mixed to describe their scrolling experience, or use Leave to exit. Source: team Cove prototype, End of time message.

UBE Data Collection

Background

The participant is a university student studying UX design. The test explored how easily the participant could understand and use Cove, including setting up a task, starting a focus session, and responding to scrolling reminders. Particular attention was given to clear navigation, understandable wording, and whether the interactions felt comfortable to use.

SUS Score: 70/100

Onboarding

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	Misunderstood “time limit” and “allowance time” as daily usage quotas rather than a period of app blocking during study.	“I thought it was a daily limit usage.”	Use clearer, consistent language to explain the app-blocking mechanism during onboarding	1

Setting Up Task

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have an upcoming assignment and want to break it down into manageable	Found task creation understandable but felt that entering subjects,	“I just want a simple task list where I can add a task, instead of feeling like I have to	Simplify task creation and make categories, subtasks and time limits	1

steps so you can stay focused and avoid procrastinating.	assignments, subtasks, dates and categories involved too many steps.	do a whole planning before studying.”	optional to reduce friction.	

Starting a Focus Room

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty Rating (1-5)**
You have an assignment to work on and want to avoid distractions while studying. Start a focus session and choose the setup that best suits your study preference.	Understood the focus-room concept and preferred starting a study session directly rather than navigating a crowded task list. Participant expected public rooms to show profile pictures or chat but questioned whether visible co-working alone would provide accountability. Participant suggested occasional progress check-ins during focus sessions.	“I’d rather start studying straight away.” “Just seeing other people there doesn’t necessarily mean they’re focusing.” “Maybe check in every 30 minutes or an hour.”	Provide a direct entry point to focus sessions without requiring users to complete extensive task planning first. Make the purpose of public rooms clearer and consider meaningful accountability cues. Offer optional, appropriately timed check-ins to support focus without creating additional distractions.	1

Pet Playground

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have completed a focus session and earned coins. Explore the Pet Playground to see how you can use your rewards to interact with or customise your pet.	Considered pet care and coin rewards insufficient motivation to complete mandatory tasks.	“It’s straightforward.” “But I feel like personally I wouldn't be motivated enough to finish my task just to earn coins for my pet at the end.”	Keep pet-related activities and reflection optional so they do not become barriers to studying.	1

Locked Out Interaction

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You are in the middle of a focus session when you feel the urge to leave Cove and open a distracting app. Follow the prompts and see how Cove responds to your attempt to leave the session.	Responded positively to the blocked-app intervention, particularly the prompt offering tips by going back to Cove which interrupted doomscrolling and encouraged them to	“Sometimes I’m scrolling because I’m stuck on something difficult.”	Use concise, supportive prompts that help users recognise avoidance and return to their task	2

	reconsider why they were avoiding work.			

Main Findings

Understanding matters beyond completion

One participant initially thought the pet was a profile picture. Another read the time-limit wording as a daily quota. The prototype does include a daily limit, so I need to confirm which version was tested before treating that interpretation as a misunderstanding. I would explain the pet through a practical example and make the difference between daily allowances and session restrictions clear. Testing should check what people understand as well as whether they finish.

Starting should require less planning

The group notes show different needs. One participant found subtasks useful for complicated assignments, while another felt there was too much setup. I would keep details available but make it easier to start with a basic task. Quick entry, optional categories and later editing would help with that. Another participant used the Add subtask prompt, which supports keeping clear action labels.

Social presence needs a clear purpose

Two participants questioned whether another avatar meant that person was focusing. A separate comment showed uncertainty about what the task checkbox meant. I would make task selection and completion look different, explain what room presence tells users, and test optional check-ins. Limiting messages might reduce distraction, but the appropriate frequency would need testing.

Interruption can help without compulsory writing

The group feedback suggests that an interruption and a reflection can feel quite different. One participant valued reassuring tips; two others wanted less writing or said they would skip it. Feedback on my earlier Intention Checkpoint also questioned the need to write an initial idea. I would test a short prompt with clear choices, optional extra writing and a direct route back to a manageable task.

The pet needs value beyond cosmetic rewards

Pet customisation did not motivate everyone. I would keep it as one way to encourage users, alongside practical help such as a reminder of their next step. There also needs to be a clear point at which users

leave the Playground. The feedback does not show that making the pet suffer or adding random rewards would address the difficulty of starting work.

Limits of the evidence

These findings come from the group report and shared interview notes. The tables above use the SUS score, difficulty ratings and feedback recorded for Participant 9. The reported group average SUS score of 84.58 describes perceived usability. It does not show whether Cove reduces scrolling or improves academic performance. Further testing should follow students' understanding, task starts and returns to work over time.

References & Statement of AI use

References

Grüning, D. J., Riedel, F., & Lorenz-Spreen, P. (2023). Directing smartphone use through the self-nudge app one sec. *Proceedings of the National Academy of Sciences, 120*(8), Article e2213114120. https://doi.org/10.1073/pnas.2213114120

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Norman, D. (2008a, November 17). *Affordances and design*. https://jnd.org/affordances-and-design/

Norman, D. (2008b, November 17). *Signifiers, not affordances*. https://jnd.org/signifiers-not-affordances/

one sec. (n.d.). *Cut your screen time in half*. https://one-sec.app/

Statement of AI use

I used ChatGPT (OpenAI) to help draft and organise tables, format references. The interview data and prototype images came from our project materials.

Jessica Zou

Task scenario

Persona: The Burnout

Name: Christie Smith

Occupation: Part-time university student

About Christie Smith

Christie is a third year university student who is burnt out, has given up on her studies and is just aiming to graduate. Recently, she hasn't been able to muster up the energy to start her work, only working on them when her friends invite her to a study date. She just wants to play games or doomscroll at home and go out with her friends, she doesn't care if it gets her a bad grade. She tried using productivity apps before but found them either not helpful or not interesting enough to continue using. She feels guilty that she is not taking her studies seriously, when her parents support her so much but it isn't always enough to make her feel motivated to study.

Behavioural considerations

- The app needs to be interesting and fun
- Needs to be more attractive to use than doomscrolling or games
- It has to be have quick learnability and be easy to use
- The user likes working with her friends
- The app needs to be able to help push her through her apathy towards her assignments

Frustrations

- She is self-aware but isn't self-motivated to do her work
- Wants to take things easy and put in little effort
- Feels guilty for not doing her work because of others
- Doesn't care about her grades

Goals

- Do her work better in a way that is easier and more fun for her
- Be able to start her assignments willingly
- Do her work with a bit more effort
- Make her parents happy and make herself feel less guilty

Tasks

- Stay focused on her work and finish it
- Keep her virtual pet happy and unlock new clothes/furniture for it
- Call with her buddies to do work together

Task Scenario- Focus Room and Pet Care

Christie Smith is a third year architecture student at the University of Melbourne. She is currently part-time and doing her capstone subject but she is burnt out from the constant work from the past 2 years and now finds it difficult to even start her assignments. She is apathetic towards this and has given up on her grades, instead deciding to prioritise things she likes like video games and hanging out with her friends. The only thing that can get her started on her work is going to study sessions with her friends and the guilt from disappointing her family.

Her friends are worried about her and suggest a productivity app they use named *Cove*. She is hesitant at first as she is not really interested in it, prior productivity apps did not interest her and didn't hold her attention but for the sake of her friends she decides to try the app out.

She is first brought to the pet selection screen and chooses a white dog, naming it Fluffy. When asked to add her friends and create a task, she skips the second one, not really caring and thinking maybe she'll add one later.

She doesn't use the app for a few days, until her friends invite her to join a focus room. She pulls up to the private room and she is relieved that it's not a super serious setting and you could send simple messages. Christie does that but her friends aren't responding, she realises that they have been studying so she stops messaging to not break their focus.

The phone blocks her from exiting the app when in the focus room, she isn't able to scroll or play games like she usually does. This combined with her friends also studying, compels her to pick up her pen and start doing her assignment.

She is able to do her work for a while before getting tired and leaving, but not before finishing a part of her work. She is able to get out of the room, relieved that she was able to finish part of her work and it elevates a bit of guilt that she feels from not doing it.

She finds the pet cute and decides to finally set some tasks to earn coins to get some outfits for it. She was already going to be studying with her friends in the focus room anyway so she sees it as just an add on to things she's already doing. Setting the task in the focus session gets her coins after every session. Over time she forms a bond with Fluffy and enjoys playing with it and giving it more accessories and new furniture.

Although, slowly integrating it into her routine, she sees this as spending more time with her friends and playing a game, her grades gradually go up without her realising it and her parents are happy too.

Design precedent

The Seinfeld Method is a method that was attributed to Jerry Seinfeld. Seinfeld believed that to improve in his craft, it's important to write jokes every day. The comedian, when giving advice to his colleague, Brad Issac, told him to get "a big wall calendar that has a whole year on one page and hang it on a prominent wall" and mark off each day and "don't break the chain" (Trapani, 2023).

This method uses the psychology of sunk-cost fallacy. Sunk-cost fallacy is a cognitive bias that once you have put an investment into something, the more likely it is for you to continue investing because you can't ignore "the cost that can't be recovered" (Nicholas, 2023). The Seinfeld method uses it in a positive manner, as a "self-managing device" and uses it to have the user commit to the activity (Hong et al., 2018), starting the chain brings more pressure to not "break the chain". The Seinfeld method has been seen to be useful in breaking and forming new habits, it has been shown in slowing alcohol consumption (Fader, 2015).

A lot of different industries follow similar philosophies showing that the method is successful. A notable example is streaks, such as in Snapchat or Duolingo used to motivate the user to keep coming back to not "break the chain". Punch cards are another example, although there is no risk of "[breaking] the chain", they incentivise the user to keep going back to see the card filled up and get their reward.

Free-to-play games in particular have evolved the formula as they have to be able to retain users to make profit. A prime example is *Project SEKAI COLORFUL STAGE! feat. Hatsune Miku*, looking at Figure 1, the game uses the incentive of rewards to strengthen the effect of

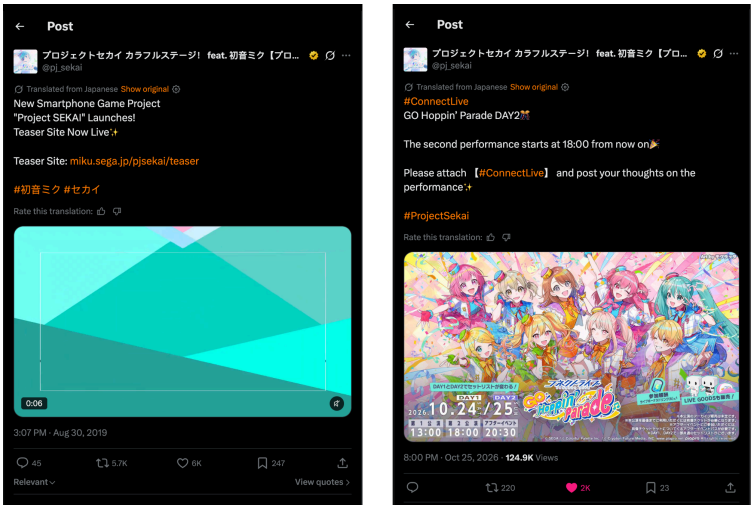
Project SEKAI weekly rewards

Note. From Project SEKAI COLORFUL STAGE! feat. Hatsune Miku app (SEGA, 2026)

the Seinfeld method, each day the user logs in and plays the game, they get a reward with a grand reward at the end of the week if they hadn't missed a day. Project SEKAI's X page has consistent likes from launch to present day as seen in Figure 2, showing the effectiveness in their player retention.

We used the Seinfeld method as inspiration in the way that our tasks function. Taking inspiration from Project Sekai, we decided to allow subtasks of the main task to each have their own due dates and rewards, ultimately culminating into a larger reward once everything is completed. Using this method, we hoped to be able to create a new habit for the user of consistent studying.

Project SEKAI X page likes comparison



Note. From Project SEKAI COLORFUL STAGE! feat. Hatsune Miku *X page* (@pj_sekai, 2026)

Usability rational

The usability principle that I focused the design around was Constraint. “Constraints limit the number of alternatives” (Norman, 2013) so that the user is able to understand how to use each functionality with minimal effort while still feeling like they have the freedom of choice.

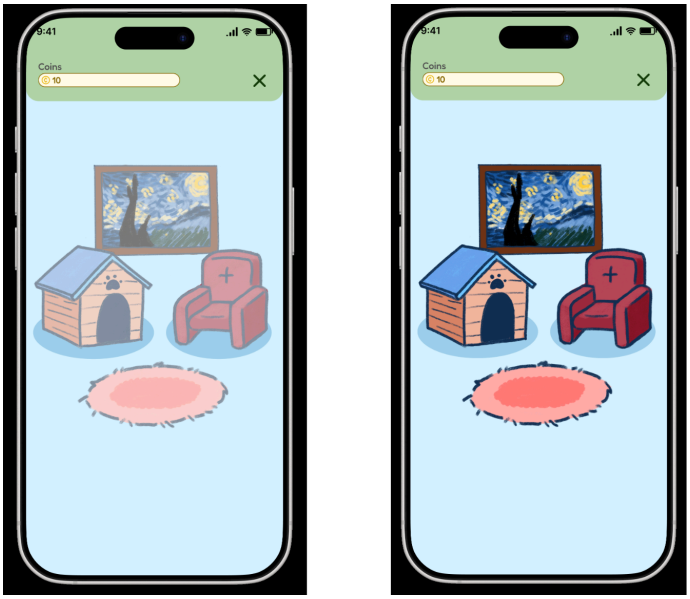
In my wireframes I have used the principle in a number of different ways to create the pet care functionality. Separating the outfits into different sections instead of presenting the user with all accessories presented at once, makes it easier for the user to understand where each part of the outfit will go as well as makes it less overwhelming for the user to scroll through to find something they like. Without this constraint, the user may not be able to understand where each of the accessories will go due to the small size on the phone screen.

The locks on the unbought items show that it isn’t open right now but can be opened under some condition, similar to how a lock works in real life which would align with the user’s mental model, making the system easier to learn and use. Similarly, the flashing furniture, seen in Figure 3, in the room customisation section aligns with the user’s mental model from other pet care apps and also differentiates the furniture from the non-interactive sections of the page, leading the user to the correct items to press.

Although I didn’t design the wireframe for the shop, in the final prototype I also helped make the shop as it is closely linked to the pet care functionality. As seen in Figure XX, when the user does not have enough coins, the item’s price turns red indicating to the user that they can't afford it. It makes it so that the user can recognise what they can afford at a simple glance, eliminating the need to look at the coins before checking the clothes or mistakenly thinking they had enough coins to buy it, reducing cognitive load.

In the user-based evaluations, the participants were able to clearly understand when items are unable to be used when asked and no one made the mistake or was confused about what can be pressed on screen. This was one of the fastest sections that the participants went through, showing that the constraints worked in allowing the user to ignore and understand what is not interactable.

Room customisation flashing



Note. From the prototype of Cove, Group 14, 2026, The University of Melbourne. (https://www.figma.com/proto/Vkt3tuczadGLkEtaFEn1zK/Cove?page-id=1%3A2&node-id=1430-7311&viewport=-9993%2C10101%2C0.7&t=xN35G34sXTce85m0-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=1396%3A5187&show-proto-sidebar=1*)*

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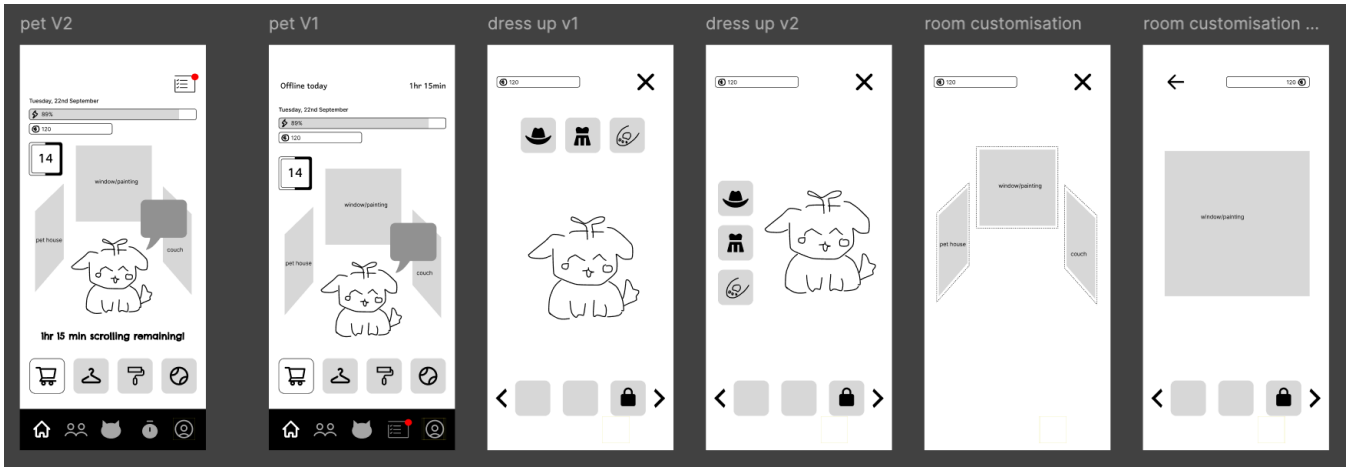
screen. This was one of the fastest sections that the participants went through, showing that the constraints worked in allowing the user to ignore and understand what is not interactable.

Wire-frames



UBE Data Collection

Participant 1

SUS Score: 87.5

Background: 22 year old university student majoring in User Experience

Onboarding

Task-Scenario	Observed Behaviour/Question asked	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating,	She goes through it very easily and understand what each step means	“Currently it's okay”	Onboarding is simple to follow and quick to get through	1 - Very Easy

so you decide to check it out and sign up.				

Setting Up Task

Task-Scenario	Observed Behaviour/Question asked	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have an assignment due soon. You write out your tasks and break the biggest one down into small, manageable steps you think you need to get started.	Was it easy to see that each little task is broken down, there's like ten coins each for like rewards?	(They look around the screen) “I didn't see the coins.”	The idea of subtasks are generally pretty good and would be useful for making tasks easier to digest and tackle. Although, there should be more of an emphasis on the rewards that could be gained for more of an incentive to add subtasks.	1 - Very Easy
	The subtasks, do you think it's useful?	“Some tasks are quite complicated and I don't see other apps doing it, so I think it's something new. It's potentially pretty good.”		

Starting a Focus Room

Task-Scenario	Observed Behaviour/Question asked	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)

You want to get into a focused headspace, so you decide to start a study session on your own.	*Was hesitating on which icon to press to go to the focus room*	“When I saw the icon, it’s more friends or just, like, I don't know, finding a buddy. Or maybe, like, chatting with friends rather than, like, a focus room.”	The focus room was generally seen as useful as it added accountability to the user but there could be more accountability added through finding ways to visualise the work that the other person is doing. This could be through displaying the task the others are doing or maybe photo check-ins. Messages were seen as unnecessary but could be included with proper moderation. There is an issue that the reflection was too long for the participant and they believed that over time it would annoy them. To solve this, we could make the reflections shorter or optional or even create a different feedback system like making it a multiple choice questionnaire which would make it easier as “recognition is easier than recall” (Cousins, 2010).	2 - Easy
	Would you study alone or with other people?	“Someone else is concentrating with me, and they can, like, keep me in check, and I can also keep them in check.”		
You suddenly remember that your friends, Jimmy and Mary wanted to study to so you invite them to your session	Asks a question - “if I change [the task], will Jimmy also see it?”	“Maybe you could make it so you both can see [the task]. I don't know. I don't know… So they know you're not, like, slacking off.”		
After a while, Jimmy has to leave but you still want someone present so you join a public room	They go through it easily			
	How do you feel about the messaging function?	“I think it might get annoying if, like, you join a bad group and they just keep spamming something. I mean, I guess to count to that you could make it so like the messages are limited to a set, like a		

		chosen set of phrases”		
After you send the session you are greeted with a reflection activity	Would you do it?	“I personally don't really like this because it's like too much yapping, too much yapping. Like, because if they like, if I end session and every time they ask this, I might get frustrated with it.” “Since you guys operate with, like, a coin system, what if you, like, it costs some coins to end this session early?” “I just prefer not a reflection, because I just hate reflection.” “I think I just find it annoying because it's compulsory. And also because. I wouldn't have much to say, like, especially if I have like 10 tasks a day I want to do.”		

Pet care

Task-Scenario	Observed Behaviour/Question asked	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	They went through each section very easily	“It's pretty straightforward.”	The pet care section was easy to use and matched the participant’s mental model.	1- Very easy
	Was it clear that some items were available and some weren't and were you able to tell what each icon does?	“Yep”		

Locked Out Interaction

Task-Scenario	Observed Behaviour/Question asked	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You set your time limits during onboarding earlier. Right now you have some scroll-time allowance left, and you open Instagram to take a short	Goes through the process pretty easily	“I know that Apple already has a feature like this… And every person I know who uses that thing never actually, like, follows through”	Overall it was pretty easy to understand but there were some pain points. Firstly, the blocking function may not be very useful because	3 - neutral

break and catch up with friends.			it is a built-in function so a user may not download Cove for it. There was a bit of confusion over the energy bar and it would be fixed through a bit of guidance earlier on in the app such as in a tutorial or guide. This would also help in other areas of functionalities as well. The tips were seen as useful and as a way to help with task initiation and the tracking of your answers was seen as neutral.	
	They didn't realise that the energy of the dog correlated to the time that they have left in the	"You could put it at the start, seeing how the energy up next to the dog is like a mini tutorial. Or like an explanation?"		
You have an assignment due in 12 hours. You open Instagram anyways, just to see what is new.	What do you think of the tips so far?	"I think it's fair. Because it gives it the choice to, like, stop.. 'Cause I think if I was really overwhelmed, I just don't want to do it, then I think this is a good selection" (of moods on the screen)		
You have marked the last 3 sessions as regretted on Instagram but you open Instagram anyways.	Do you think it's invasive?	"No. But I just don't really care about, like, privacy shit, so, my comment might not be, like, the general audiences"		

Other Interactions

Observed Behaviour/Question asked	**Participant Feedback**	**Usability Insight**

Do you think this app will be useful to stop from scrolling and procrastinating?	“I've used similar apps before, and it's, like, it only works for, like, a week, and then it just doesn't work afterwards, for me personally.”	The participant had a couple suggestions for us at the end, mostly for user retention on the app. They suggested a gacha system instead of a shop to stand out against other apps, it would give the users a reason to get a lot of coins and combining it with a rarity system would heighten the dopamine gained from the thrill of not knowing when the reward comes (Anselme & Robinson, 2013). They believe that a good aesthetic and theme would really improve the draw of the app.
What would make you use the app?	“Oh, um, like, definitely the coins, would encourage me to keep going, especially if the customisation is, like, really good, then I'm, like, yeah, I need that costume. or something.”	
Which part would you think is the most useful?	“I guess the coin system, because the pop up and Instagram might just annoy me more than anything. because I've tried something similar and, like, I low key just deleted the app. after they popped up enough times.”	
Is there anything you would change?	“I don't even know if this is feasible, but what if the coins are for, like, a gacha system?” “I feel like because the study apps don't really do gacha	

	systems, I think it will be really interesting if one does try it. I'll definitely try it out myself." "I think aesthetics go a long way, so I would only recommend it if it looks good…I feel like it would be more interesting if the pets weren't regular pets... Could be like an alien or something."	

Participant 6

SUS Score: 92.5

Background: 22 year old university student majoring in Animation

Onboarding

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty Rating (1-5)**
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	Goes through the onboarding process with ease and efficiency		The onboarding was easy to understand and the participant had nothing they would change about it.	1 - Very Easy

Setting Up Task

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have an assignment due soon. You write out your tasks and break the biggest one down into small, manageable steps you think you need to get started.	Was it easy to see that each little task is broken down, there's like ten coins each for like rewards?	“Yes, but I don’t really know what the coins even do right now”	The participant was able to go through the task creation process easily but was confused about what the coins were even for. To solve for this we should add more context at the start of the user experience, such as in a tutorial.	1 - Very Easy

Starting a focus room

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You want to get into a focused headspace, so you decide to start a study session on your own.	Wasn’t sure when choosing the task if	“I thought it was me accounting for the task during the session, but then when I ticked it, it came up with the plus 10, so I was like, oh, maybe I'm saying I	The focus room was pretty easy in general but there was part that didn’t match the participant’s mental model. The participant is confused by the “plus 10” part of the task list, the easy solution is just remove it and include it either in	2 - Easy

		did the task because I ticked it.”	the tutorial or in an information box. They were very into the idea of having company around even virtually being able to help with personal accountability. They received the functionality because you got to control the length of the writing so if someone didn’t want to do it they could just type something random.	
You suddenly remember that your friends, Jimmy and Mary wanted to study to so you invite them to your session	*Would you study alone or with other people?*	“I like that my friends are also doing it, like it's nice so like all come together or even if they're doing different things, it’s also kind of nice but then sometimes I guess I'd want to study by myself, depending on, like, my mood, I guess. Yeah.”		
After a while, Jimmy has to leave but you still want someone present so you join a public room	Would you be hesitant to join a public room?	“I feel like it's kind of like a nice, passive company to know other people have to lock in as well. So I don't think I would hesitate. Although, I would be more inclined to join my friends if they were there.”		
After you send the session you are greeted with a reflection activity	It wouldn't be too annoying to have that every time you finish?	“I think it's cool. Depending on, like, my mood, I guess sometimes, like, it would be, like, a longer or shorter reflection. But I think it's a cool feature.”		

		“I feel like it's not, like, that annoying if it's mandatory because it's not like you have like a minimum amount of like characters.”		

Pet care

Task-Scenario	Observed Behaviour/Question asked	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	The were very quick at this section, wasting no time	“Oh, my gosh. This is so cute.” “That's awesome. I'm locked in.”	They thought the pets were cute and went through the section very easily with no trouble.	1- Very easy

Locked Out Interaction

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)

You set your time limits during onboarding earlier. Right now you have some scroll-time allowance left, and you open Instagram to take a short break and catch up with friends.	So you've reached the end of your allowance and it asks you to write how you felt from scrolling, how good of a time you felt. Do you think that that's like a good thing to ask, not too much pressure?	“It makes you like think about like, what you're actually spending your time on, because I feel like if you're doing something that felt like enjoyable to you, like watching a movie or something at the end, I feel like I don't usually feel like, “ah, shouldn't have watched that movie”. But like when you're scrolling, sometimes I feel like you're like, “yeah, that was just like a waste of time.” So it's good to think about, Like if you did have like a relaxed time scrolling, because sometimes I have a little scroll and I'm like, “ah, that was a funny joke that that person did,” but then sometimes you're like, “uhh,”	The participant has a tendency for avoidance and the app physically stopping them from continuing is really helpful for them. It helps them think twice before scrolling when they might regret it. They didn’t find anything. They really have no complaints about the feature but they thought it might be tempting to lie about your feelings when entering the app. There is no direct solution to this but using softer and more gentle language might help them feel less like they need to lie about it.	1- Very easy
	Does it make sense for it to exist like that? And if you were stuck on a task, do you think you would be truthful about it?	“Yeah, I think it makes sense for it to ask that. Would I be truthful about it? I guess it depends. I'd want to say that I'd be truthful about it. But um. I guess it depends. If I was just like in the moment, “ah, whatever, I just want to scroll”. Like maybe I'd be like, ah, I'm just taking a		

		break. But, like, I feel like if I'm using the app to like focus, I'd want to have a sense of integrity about it”		
	What makes more sense for you, having different time limits for each app or having a unified time time limit just for blocked apps in general?	“I think having a unified time makes more sense… A specific time to set it would feel like it would feel better if I had like one time limit and could choose, I guess. And then I wouldn't have to decide, like, per app, it just feels more concise”		
You have an assignment due in 12 hours. You open Instagram anyways, just to see what is new.	Do you ever feel like you would go on the app because you're stuck on the task? Like, do you think that's an appropriate question to ask?	“Yeah, stuck is a good question to ask because I feel like if the task felt comfortable to do, I probably wouldn't leave it, to like scroll if the task was like something that I felt like I knew how to do.”		

	Do you think the tips are useful?	“Yes, because sometimes I don’t want to do my work and I feel like this might help.”		
You have marked the last 3 sessions as regretted on Instagram but you open Instagram anyways.	Do you think it’s invasive?	“I'm, like, not a super security conscious person. So maybe for someone who's like, who cares a lot about security, they would be like, “I don't know if I want it to monitor me.” But for me, I feel like it's something I'm using to be helpful to me so the data is like just something that's helpful to me.”		

Other Interactions

Observed Behaviour/Question asked	**Participant Feedback**	**Usability Insight**
Do you think this app will be useful to stop from scrolling and procrastinating?	“I feel like you're keeping each other accountable, like kind of.”	Overall this participant focused a bit more on the accountability aspects, seeing it as one of its strengths, which is why they mention the focus room quite a bit. The participant has had a history of using productivity apps and the one that worked for them was a blocking app so that was one type of app that is proven to work for them.

What would make you use the app?	“I guess it would depend on, like, I guess I feel like if I wanted to improve my productivity, that only reason I'd like, stop using it is if like, I didn't really have anything to study,”	
Which part would you think is the most useful?	“Probably like stopping from doom scrolling, I think the limits would be really good for that. And then I think about helping to study, I like the focus room and like joining rooms and starting rooms most. And like that you can add tasks in there. I think that's good.”	

Main Findings

The main findings from the two usability tests conducted was that the pet care and focus room are the parts that they find the most useful. Both participants think that the focus room is useful due to the accountability that comes with working with other people. There are a bunch of small issues with it but overall it was received positively. Although it is interesting to note that looking at interviews that other group members did, pet care was not that popular of a functionality for some of the other participants, leading me to hypothesise that pet care is more popular amongst art and design students. Tips were seen as useful from both people but Participant 1 does think the enough lock out function would annoy them,

The main addition to the app that would help elevate many of the participants’ troubles is to add a tutorial or guide after onboarding that highlights all the key features, this will help prevent confusion down the line. There could also be info sections implemented in features that frequently cause confusion. Another

change could be to improve the theming of our app because the lock out function and the pet care function do not have too much visual connection right now.

References & Statement of AI use

Acknowledgement of tools used: I have not used any AI tools or technologies to prepare this assessment.

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Jiexi Xiao

Task scenario

Persona: The Last-Minute Starter

Name: Ethan Chen

Age: 21

Occupation: University student with a part-time job

About Ethan Chen

Ethan is a third-year university student who works part-time alongside his studies. He cares about his grades, but usually leaves assignments until the deadline is close. If something is due next week, he assumes he has plenty of time and puts it aside. Work shifts and other assignments then fill up the week, and he ends up rushing. He has tried keeping lists in his phone notes, but often forgets to check them. He wants to be more organised, although he does not want planning his work to become another job.

Behavioural considerations

• He is more likely to start when he can see a specific action rather than a whole assignment.

• Adding a task needs to be quick, with extra details left optional.

• Deadlines should be visible without opening every task.

• His schedule changes around work, so he needs to adjust unfinished tasks.

• Progress should stay visible when he returns after a break.

Frustrations

• He underestimates how long assignments will take.

• He spends time on easier tasks while putting off urgent ones.

• He forgets what he planned to do when he returns to an assignment.

• Missing a planned date makes his task list feel out of date.

Goals

• Start assignments before the last evening.

• Know what needs his attention first.

• Fit study around his shifts.

• Leave enough time to check his work before submitting.

Tasks

• Add assignments and their deadlines.

• Break larger assignments into a few manageable steps.

• Choose a task and start a focus session.

• Record completed steps and reschedule work that is still unfinished.

Task Scenario - Tasks and Starting a Focus Session

Ethan gets home from work on Wednesday evening and opens the shared document for a group report due on Friday. His section is still empty. He has been putting it off all week, and now he cannot quite remember what he needs to include. He opens Cove, adds the report with Friday's deadline and breaks it into three steps: check the requirements, draft his section and check the references. Reading the brief feels manageable, so he chooses that as his first step and starts a private 30-minute focus session. An unclear instruction soon sends him to the group chat, where he asks for help. While waiting for a reply, he opens Instagram out of habit, but Cove's prompt makes him stop and reconsider. He returns to the document, leaves the question for his group and works on the parts he understands. By the end of the session, he has a rough outline. He ticks off the first step, receives ten coins and renames the next one "Draft the research findings" so he knows where to pick up.

The next day, Ethan is asked to start work earlier than expected. With less study time available, he moves the drafting ahead of a less urgent task and works on it before his shift. He sends his writing to the group on Friday, although the references still need checking. When he next opens Cove, the task is overdue. Rather than mark the whole report as finished, he reschedules the reference check before the group's final submission. The steps he has already completed stay ticked, leaving him with one clear job to return to.

Design precedent

My precedent for Cove's Tasks section is the coffee shop stamp card. Each purchase adds a stamp, so customers can see what they have collected and how far they are from a reward. This relates to a problem with assignments: students can spend hours working without feeling they have finished anything. I wanted the task list to show that smaller pieces of work still count, even when the assignment itself is incomplete.

Nunes and Drèze (2006) examined this in a study using car-wash loyalty cards. One group received an empty card requiring eight stamps, while another received a ten-stamp card with two stamps already filled. Both groups needed eight purchases, but customers with the partly completed card were more likely to finish. Their findings suggest that feeling a goal is underway can encourage people to continue. Cove does not award progress before work is done. Instead, completed subtasks stay visible, giving students a record of where they have already made a start.

The card alone, however, does not explain how to make an assignment easier to begin. Bandura and Schunk (1981) found that children given short-term learning goals developed stronger mathematical skills, self-efficacy and interest than those given distant goals or no specific goals. The study involved children rather than university students, but it helped inform the use of smaller steps in Cove. A student can turn "Finish the report" into something they can attempt now, such as "Read the assignment brief". This remains optional, since breaking down a simple task could create more work than it saves.

Harkin et al. (2016) provide further support for recording progress. Their review of 138 studies found that interventions encouraging progress monitoring improved goal attainment on average. In Cove, checkboxes, a completed-step count and a progress bar show what has been done. Completing a subtask also gives the user ten coins. Like collecting a stamp, this offers a small acknowledgement before the larger task is finished.

Unlike buying coffee, completing subtasks involves different amounts of effort. Two finished steps out of five do not necessarily mean forty per cent of the assignment is done. The display therefore records completed steps rather than claiming to measure productivity. Students can also move unfinished work to

another date without losing their earlier progress. For someone who regularly starts late, the aim is to make the next action easier to find and give them a reason to return.

Usability rational - Feedback

Feedback helps users understand what happened after an action and whether it had the intended result. Shneiderman (1983) describes immediate, visible responses as an important part of direct manipulation. This informed the design of Cove's Tasks section, where students need to know whether their choices have registered and how their work is progressing. A task list should do more than hold assignment names; it should help users see what has changed when they return.

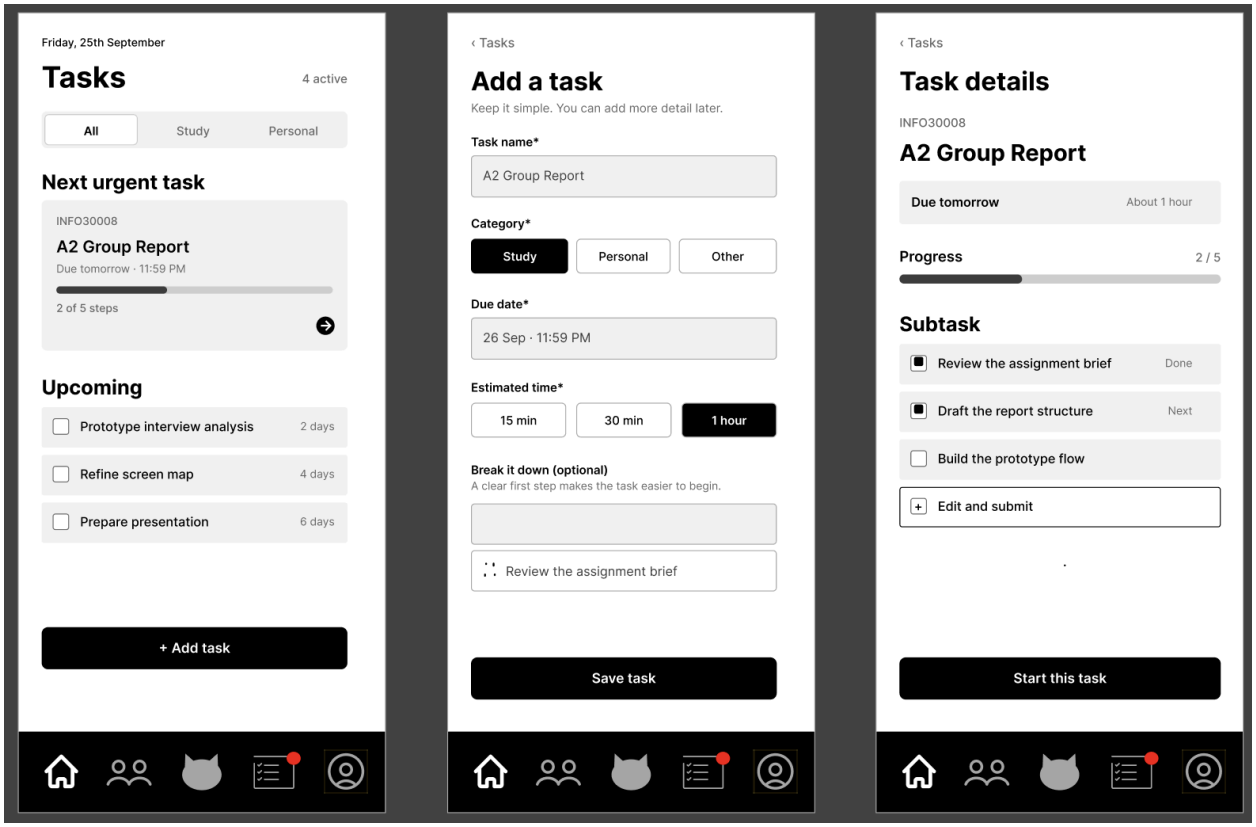
In the task creation screen, selected categories and time estimates use a dark background, while the other options remain white. This confirms the current selection before the user saves. The same approach applies to the category filters on the task list, so users can tell which group of tasks they are viewing. After saving, the new task should appear in the list with the details they entered. Seeing the task provides confirmation that it was saved, rather than leaving users wondering whether they need to enter it again.

The progress bar and completed-step count provide feedback during the assignment. When a student marks a subtask complete, the checkbox, count and bar should update together. Harkin et al. (2016) found that interventions encouraging progress monitoring improved goal attainment on average. In Cove, these indicators let students see that work has been recorded even when the overall assignment remains unfinished. The count also explains the bar: "2 of 5 steps" is more specific than an unlabelled visual indicator. It records completed steps without suggesting that every step requires equal effort.

Feedback also needs to help users decide what to do next. Hattie and Timperley (2007) discuss feedback in relation to goals, current progress and next steps. Although their work concerns educational feedback, this distinction is useful when considering the task details screen. The assignment title identifies the goal, the progress display shows completed work, and the subtask list provides possible next actions. The "Start this task" button then gives users a way to move from planning into a focus session.

The wireframe still needs some refinement. One subtask is labelled "Next" but has the same filled checkbox as the row labelled "Done". This could make users think both are complete. Only finished steps should appear checked, with the next step identified separately. These responses would need testing in the interactive prototype, since a static wireframe cannot establish whether users notice or correctly interpret a change.

Wire-frames



UBE Data Collection

Participant 10

SUS Score: 92.5

Background: Fictional 21-year-old university student working part-time

Onboarding

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend recommends Cove. Complete the initial setup.	*Paused at the allowance setting and reread the description before continuing.*	*Fictional quote: "Wait, is that how long I can scroll, or how long it blocks Instagram?"*	*A possible issue to test is whether scrolling allowance and focus-session duration are clearly distinguished.*	*2*

Setting Up Task

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
Add an upcoming assignment and organise it into smaller steps.	*Entered a task name and deadline, then looked for Save before filling in the other details.*	*Fictional quote: "I'd probably just put the deadline in now and sort the rest out later."*	*Check whether users can save quickly and recognise which fields are optional.*	*2*

Starting a Focus Room

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty

				Rating (1-5)
Start a focus session using the setup that suits you.	*Chose a private room and 30 minutes, then checked that the selected task carried through.*	*Fictional quote: "Yeah, thirty minutes is fine. I just need to actually start."*	*Keep the chosen task visible when entering a focus session.*	*1*

Pet Playground

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
Explore how coins earned from a focus session can be used in the Pet Playground.	*Understood the shop but returned to Tasks without selecting an item.*	*Fictional quote: "It's cute. I'd have a look, but I wouldn't study just to buy it clothes."*	*Test pet rewards as optional encouragement rather than assuming they motivate every user.*	*1*

Locked Out Interaction

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)

During a focus session, open a distracting app and respond to Cove's prompts.	*Read the initial reminder but hesitated when asked for a written response.*	*Fictional quote: "The reminder's fair, but I don't wanna write something every time I open it."*	*Investigate whether repeated reflection becomes frustrating and whether a shorter response would help.*	*1*

Main Findings

This example highlights areas to explore in future testing: how quickly users can add a task, whether they understand the time allowance, and how they respond to repeated reflection prompts. These points have not yet been confirmed through user testing.

References & Statement of AI use

I acknowledge using ChatGPT to assist with research and to explore and refine ideas for this assessment. I independently checked the suggested sources for credibility and relevance. All final decisions and interpretations are my own.

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Maryam Al Mahdi

Task scenario

Task scenario, with persona: the Dual Role Creator

Name: Jennie
Age: 21
Occupation: University student and part-time social media manager for a small brand
Quote: “I can’t just block Instagram, that’s where my clients message me.”

About: Jennie uses Instagram for work and for distraction. She posts content, replies to DMs and checks analytics for brand collaborations. Blocking the app completely would cut her off from her income. She’s usually well organised, but the feed is one tap away from every work task, so it’s always tempting.

Goals

- Keep work use separate from scrolling
- Stay accountable without losing access to the features she needs for work
- Finish a specific task on the app without drifting into unrelated content

Frustrations

- Can’t use blanket blockers because the app is also where paid work happens
- A quick work task happens on the same screen where endless scrolling starts
- Existing tools treat all app use as bad, so they block everything or nothing
- Loses track of time even in sessions that started as work

Behavioural considerations

- Needs control over her use rather than a full block
- Opens the app with a work intention that's gone within seconds
- Would drop any tool that blocks professional features
- Responds better to visible friction and accountability than to restriction

Why this persona matters

Our problem domain is young adults who struggle to act on their intentions when long-term goals compete with immediate digital rewards. Jennie is a useful case because her obligation and her reward sit inside the same app. Most screen-time tools assume less access is the fix. Even daily limits don't fit her, because her work doesn't stop when her allowance runs out.

Her case shows that access isn't what goes wrong for her. Drift is. She opens Instagram with a clear purpose and loses it as soon as the feed loads. So the design challenge is to hold onto her intention at the moment it's most likely to slip, while still letting her into the app.

Task scenario

It's 4:40pm on a Thursday and Jennie is in the library working on an assignment. Her phone lights up. The marketing coordinator for the brand she manages needs her to approve a caption and story mockup before the campaign goes live at 6pm. She knows a quick check usually turns into forty minutes.

So she opens Idle instead of going straight to Instagram. She sets her ticket, "Reply to brand DM", and tags it as work. Because it's a work ticket, Idle opens Instagram straight into her messages instead of the home feed. A thin bar at the top of the screen shows her ticket and nothing else.

She checks the mockup, approves the caption and replies to the coordinator in under two minutes. The bar turns green and changes to "Ticket done. Anything else for work?" with two options: add another work ticket, or leave.

She taps the Instagram logo instead, out of habit, and the home feed loads. A friend's story is lit up and a reel starts playing. Nothing blocks her, but the top bar now reads "Off-ticket - 1 min", and the minutes keep counting. Seeing her work session turn into scrolling time is enough to make her stop. She closes Instagram and goes back to Idle. Her session shows two minutes on the ticket and three minutes off it, and her pet, Milo, is sitting next to the finished ticket.

Jennie goes back to her assignment. That evening she wants to see how the campaign is doing, so she opens Idle again and sets a new ticket: "Check campaign analytics." Idle suits her because every visit starts with a reason, and she can come back as often as her work needs.

What this scenario contributes

I wrote this scenario for Idle, our concept before Cove. It still carries Cove's central idea: respond to the user at the moment they're tempted instead of punishing them for being distracted. It also brings in another stakeholder. The brand coordinator relies on Jennie replying quickly, so the design has to protect her work access as carefully as it limits her drifting.

It suggests functions beyond our final prototype too. Automatic task detection through a platform's messaging tools would let the app recognise finished work without asking. Tickets could be scheduled around client deadlines, and a work tag could carry across apps such as email or Slack. These raise open questions: the off-ticket timer needs to be read as a nudge rather than a telling-off, and the app has to tell work use from drift without feeling intrusive.

Design precedent

O3 App:

The O3 app was built for O3's 24/7 study cafes, but its main features work anywhere (O3 Space, n.d.). Its Pomodoro timer blocks distracting apps for the length of a session. Finishing a session gives the student a chance to catch an "Oyster", and the longer they study, the higher the catch rate. Studying with friends, even from home, raises everyone's catch rate, and because each person discovers different Oysters, friends help complete each other's collections. Some Oysters unlock real discounts at the cafe.

The key innovation is that O3 turns a study timer into something social and collectible. O3 gives a small, unpredictable reward, and studying alongside friends improves the odds for the whole group. This draws on social facilitation, where people tend to work harder on familiar tasks when others are present (Zajonc, 1965). O3 makes that presence pay off.

I designed Cove's Focus Room and exit reflection, and O3 shaped both. Like O3, a Focus Room lets friends study together from anywhere, runs a timer and keeps distracting apps blocked. Even the break screen tells students "Blocking stays on". Students can study alone, invite friends with a room code or join a public room, where each person appears as their pet. O3's app has an open live chat for finding study buddies which I deliberately left out. Cove's Focus Rooms only allow short messages like "Lock in!", so students get the feeling of company getting carried away in threads of messages. O3 also shows where my design could go further. In O3, friends change your outcome. In my Focus Room, friends are visible but don't affect what you earn. A shared room bonus, such as extra coins when everyone finishes their task, would bring in O3's idea of making the group's effort count together. Presence alone may not be

appealing enough either. Two participants in our evaluation questioned whether another person's avatar meant they were actually focusing, and one suggested showing everyone's task. My "You are working on: [task]" label could appear under every pet, not just the user's own.

The exit reflection is another area where I deliberately moved away from O3. O3 rewards time: a longer session means a better chance at a catch. When a student ends a Cove session, they're asked whether they finished their task. "Yes, reflect & end" leads to a short reflection and then coins. "No, but exit anyway" ends with zero coins and a nudge to get back on track. In Cove, the reward follows finished work, not minutes on a timer. Students end the session when they feel they've done what they set out to do, not when a timer tells them to.

References

O3 Space. (n.d.). *O3 app*. Retrieved October 25, 2026, from https://o3space.org/app

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Figure 4. *O3 app study session. Each friend who joins adds +5% to the shared catch rate for the Oyster ("Oliver", 45%). Source: O3 Space app, https://o3space.org/app*

Usability rational

User Control & Freedom

User control and freedom is one of Nielsen's (1994) usability heuristics, focused on minimising the extent to which a system "traps the user in a specific state". Users often choose functions by mistake and need a clearly marked "emergency exit" to leave an unwanted state (Nielsen, 1994)

This principle matters more for Cove, as cove deliberately adds friction first. If that friction also feels like a trap, students will resent it and delete the app. I used the principle so that every moment of friction in the Focus Room comes with a clear way out.

The exit check is the clearest example. When a student taps End Session, they choose between "Yes, reflect & end", "No! Stay in session" and No, but exit anyway. No path is hidden, and a student who opened the menu by mistake can go straight back to work. "Exit anyway" still works but it ends with zero coins. The student stays in control, and the cost of leaving early is visible.

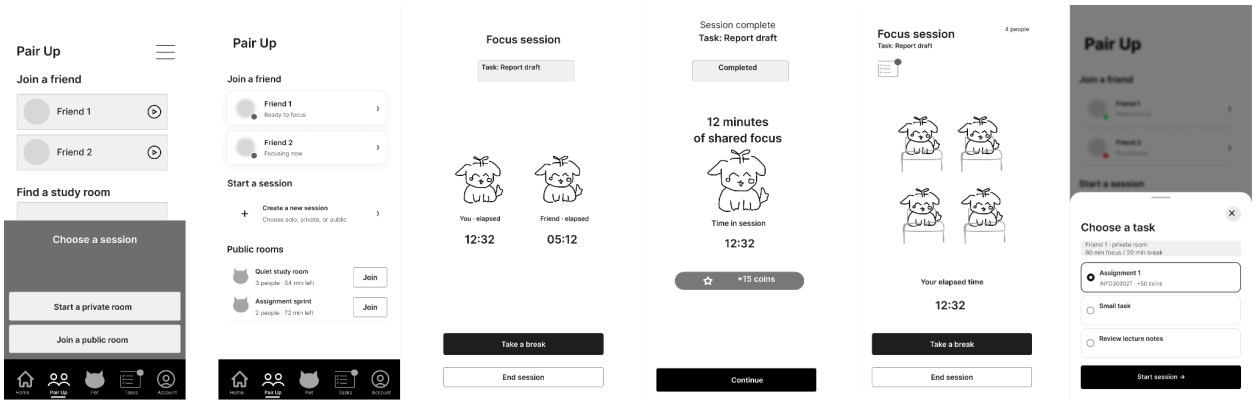
Breaks for the focus session follow the same idea. "Take a break" pauses the session without ending it, and "Resume session" returns the user to the same room and task. The break screen tells them "Blocking stays on," so they know what the pause does and does not change.

The popup when starting a room has cancel and close buttons, so starting a room is not an irreversible commitment.

Freedom also covers how students study. They can work alone, invite friends with a room code or join a public room. The session timer counts up rather than down, so Cove never ends a session for the student or pressures them to rush. They decide when the work is done. However, P7 expected the session to end when time ran out, so the count-up needs to be clearer on screen.

My evaluation showed where control was unclear. P7 asked, "Do I need to invite friends?", which suggests the prominent Invite Friends button made a free choice look required. I would label te private room as a complete solo session. P7 also dismissed both locked-out warnings and then asked "Why is it letting me go and scroll?", showering the limit of this principle in Cove: too much freedom at the wrong moment may work against what the user wants. The exit check tries to balance this by never blocking exit, while making the student still pause and reflect, before really exiting.

Wire-frames



UBE Data Collection

<u>Participant 7</u>

SUS Score: 72.5/100

Background: 18-year-old final-year high school student who admits to doomscrolling and sees it as unproductive (“I don’t ever think my doomscrolling is productive”). Comfortable with apps and studies with friends.

Onboarding

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You’ve been meaning to cut down on	Read the first screen as choosing a profile picture, then choose a companion.	“Let me choose my companion... I think I want this cat. It’s super cute.”	The companion’s purpose isn’t clear at first glance. A short line from the pet explaining what it is would stop it being	2 - Easy

procrastinating, so you decide to check it out and sign up.	Correctly identified the app-blocking step, including the energy drain. Reacted positively to adding friends and completed the task setup (category, due date, estimated time)	"So this lets me choose which apps I get to block." "I can even invite friends. How cool is that?...	read as a profile picture.	

Setting Up Task

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty Rating (1-5)**
You have an upcoming assignment and want to break it down into manageable steps so you can stay focused and avoid procrastinating.	Created a main task and a subtask without help, using the "Add subtask" prompt. Questioned why tasks need a category, then saw the reason.	On adding categories: "Why do I need to do that?... Oh, for myself, yes."	Users' behaviour shows that the adding task flow followed standard conventions. The "Add subtask" label worked. The category field should explain its purpose where it's asked, or be optional.	2 - Easy

Starting a Focus Room

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have an assignment to work on and want to avoid distractions while studying. Start a focus session and choose the setup that best suits your study preference.	Asked whether they needed to invite friends, then found the Focus Room and started a private room. Chose a task, took a break and resumed. At the end, went through the completion check and understood the coin reward. Seemed to expect the session to end when time ran out, but it isn't a timer.	"Do I need to invite friends?" "So now I get to spoil my pet with some of the coins earned, yay."	Studying solo should read as a valid choice, not a missing step. "Time in session" was read as a countdown, so it needs to show clearly that it counts up and gives user flexibility for how long they'd like their session to be. The completion check and coin reward were understood and well received.	2- Easy

Locked Out Interaction

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You are in the middle of a focus session when you feel the urge to leave Cove and open a distracting app.	Read the midpoint and 5-minute warnings aloud and understood them, but dismissed both and	"No, let's continue on the app." "It was regretted, I don't ever think my	The warnings were clear but easy to dismiss. The participant regretted the scroll yet was still allowed to continue, so a firmer	1- Very Easy

Follow the prompts and see how Cove responds to your attempt to leave the session.	kept scrolling. Asked to rate their scrolling, called it regretted and questioned why the app let them keep scrolling.	doomscrolling is productive.” “Why is it letting me go and scroll?”	checkpoint near the end of the allowance could help.	

Main Findings

P7 completed all four scenarios they tested, rated each one easy or very easy, and gave Cove a SUS score of 72.5, slightly above the benchmark of 68. The interesting findings are less about whether they could finish tasks and more about what they expected the app to do.

Finishing a step didn’t always mean understanding it

P7 first read the companion screen as choosing a profile picture. In the Focus Room they expected the session to end when the time ran out, although “Time in session” counts up. Both screens were completed without help, but the participant’s mental model was different from ours. A short line from the pet explaining what it is, and a clearer label on the session time, would close these gaps.

Studying alone needs to look like a real option

Before starting, P7 asked, “Do I need to invite friends?” The Focus Room shows a filled “Invite Friends” button next to the session controls, which may have made inviting look like a required step. As the designer of this feature, I would make the private room’s purpose clearer on entry so a solo session reads as a complete choice.

Tying the reward to finishing worked

P7 went through the completion check, understood why they were getting coins, and reacted happily (“So now I get to spoil my pet with some of the coins you earned, yay”). This supports my decision to reward finished tasks rather than time spent.

The locked-out prompts were understood but too easy to ignore

P7 read both warnings aloud and dismissed both to keep scrolling. When asked to rate the scroll, they called it regretted and added, “I don’t ever think my doomscrolling is productive.” They then asked, “Why is it letting me go and scroll?” This is the gap between intention and action that our problem domain describes: the participant knew the scroll wasn’t helping them and still continued. The warnings were rated very easy, which may mean they were easy in the wrong way. A firmer checkpoint near the end of the allowance is worth testing. Other participants in our group found frequent interruptions annoying, though, so the balance needs care.

Limits of this evidence

These findings come from one participant, a final-year high school student rather than a university student. The facilitator prompted the friend to invite and nudged the locked-out response, so those moments weren’t fully spontaneous. Pet Playground wasn’t tested in this session. P7 is a friend, so the SUS score is likely optimistic, and the observed behaviour should carry more weight than the ratings.

References & Statement of AI use

AI (anthropic) was used to help draft and edit the written parts of my individual section. I also used it to find and check resources, format references and trim sections to the word limits. The design ideas, prototype screens, and participant data and quotes came from my own work and testing sessions. I reviewed and edited all AI-assisted text before including it.

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Rumaysa Fatima

Task scenario

The Overwhelmed Avoider

Persona:

Michelle Choi is a 20-year-old university student who procrastinates selectively on tasks that feel emotionally heavy rather than simply time-consuming. These are things such as essays, unfamiliar software, and anything that feels quite uncertain to her. However, when tasks are simple and well-defined tasks they get done without a fight. This selectivity is important to her design relevance as Michelle is not lacking in general motivation or time-management skill, which means tools built around scheduling, reminders, or willpower miss the actual barrier she faces.

What Michelle avoids is not the effort of the task itself but the feeling of dread and fear that precedes starting it. She does not fully trust her own ability to do the work well, and rather than risk a poor attempt, she avoids beginning at all. This is a pattern that produces a self-reinforcing cycle of guilt over not starting which in turn feeds further avoidance, feeding more stress, making starting feel even harder. Her tolerance for disruption is thin and this means that a single unexpected setback, such as a crashing program or a confusingly worded brief, is often enough to send her back to scrolling rather than troubleshooting through it.

This shapes two clear frustrations with existing tools. First, productivity apps generally assume the missing piece is a schedule, when Michelle already knows what she needs to do, meaning Michelle's gap is how to begin, not when. Second, tools that lean on pressure-based framing, such as prominent due-date counters or “you’re falling behind” messaging, actively work against her, since they amplify the exact anxiety she is trying to escape rather than lowering it.

Michelle’s goal is that she wants her tasks to feel less overwhelming, met with interventions that match the size of the task to the size of the step she is actually capable of taking right now.

Task scenario:

It’s 6:00pm on a Tuesday. Michelle has an essay due at 7:00pm in four days. This is enough time that there’s no immediate crisis, but close enough that she knows, with the familiar sinking certainty she’s felt

before, that if she doesn't start properly tonight, she'll regret it later. The problem is that she barely understands the topic. She's read the brief twice and still isn't sure where to begin, and that uncertainty is exactly the kind of fog that usually sends her reaching for her phone instead.

She opens Instagram out of habit, more than intention. Cove recognises that she has a task, "Essay: Topic X", sitting in her to-do list with a due date four days out, and rather than a blunt countdown or a guilt-inducing "you're behind" banner, it simply asks: "What's going on right now?" Michelle taps "I'm stuck with my task." This matters to her specifically, because the app isn't assuming laziness or demanding justification, it's treating the moment as a stall to work through, not a failure to be accounted for.

The next screen asks which situation fits best, and Michelle selects "I'm feeling overwhelmed." Importantly, this doesn't funnel her into a generic productivity tip. The response names her exact experience back to her saying "It's ok to feel overwhelmed about our work sometimes", before offering something concrete. Cove suggests she makes the start smaller, with three plain steps of firstly, opening the task outline, then, adding one bullet point under the first heading, and finally a stop there if that's enough for now. There is no mention of the due date, no progress percentage, no pressure framing of any kind and this is precisely the kind of design Michelle's frustrations explicitly warn against.

Michelle taps through to her task list. She opens the outline for her essay and, rather than feeling the usual pull to have a fully-formed plan before she's allowed to "really" start, she writes one bullet point under her first heading which is just a rough note on what the introduction might cover. It isn't perfect and it doesn't need to be. The instruction told her to stop there if that was enough, and for tonight, it is.

She closes the task feeling relief rather than dread she usually has for this stage of an assignment. The guilt-avoidance-stress cycle that would normally have carried her straight back to scrolling hasn't had the chance to take hold, because the task was broken down before she could escape to her avoidant habits. Three days later, with that first bullet point already sitting in the document, starting her second session on the essay feels measurably less daunting, and this is not because the essay itself got easier, but because she's no longer starting from zero, and the fear of the blank page, her actual barrier, had already been addressed before it could stop her.

Design precedent

Pointing and calling (Shisa Kanko) is a method which originated in Japan in the early 1900s as an error-prevention routine. The worker looks at an object, points at it, and says aloud what they are confirming. It started when a steam train driver worried about failing eyesight had his fireman call back

what he saw (Flight Safety Australia, n.d.). By 1913 this call-and-response was written into a railway manual, and pointing was probably added after 1925 (Japan Times, 2008).

Many variations of this technique exist. For example, Aviation flight-deck procedures typically use pointing or calling, rarely both (Flight Safety Australia, n.d.). Additionally, New York's su

bway began requiring conductors to point at platform "zebra boards" in 1996 (AOPA, 2021). In healthcare, nurses who pointed were better at selecting the right drug among similarly named ones (Pointing and Calling in Healthcare, n.d.), and a Canadian team now teaches it to paramedic students (NAIT, 2025).

There are many different examples to show how successful this method can be in different contexts. A 1994 Railway Technical Research Institute experiment found the lowest error rate, 0.38 per 100 actions, when volunteers both pointed and called, an almost 85% reduction (Japan Times, 2008). An Osaka University study concluded the technique supports working memory and the response that follows (Code7700, n.d.). The evidence has limits: the best figures come from simple lab tasks, and some Japanese workers see no point in it, which is why training machines exist (VoiceTube, n.d.).

Pointing and calling does not stop an action or punish a mistake and this is where its key design concept sits. It inserts a brief, externalised declaration at the moment of action, turning autopilot into a conscious choice. Researchers expected it to improve accuracy but slow people down (Code7700, n.d.), yet errors virtually disappeared. This resolves a paradox Cove shares which is how to interrupt impulsive behaviour without removing temptation or adding punishment.

This can inform the decisions made in Cove’s design in many ways. Firstly, when the user is tempted to use a blocked app while having some allowed time, the friction points asks them to state what they are about to do and why. That is the "call" at the moment of action, not a block. Moreover, The Japanese Times (2008) state that pointing and calling work best together. Likewise, Cove pairs a physical action, which is a typed reflection before entering, with a stated reason, not a dismissible pop-up.

Usability rational

Maintain Consistency and Adhere to Standards (Usability Heuristic #4)

Nielsen's fourth heuristic holds that users should never have to wonder whether different words, situations, or actions mean the same thing, and that interfaces should follow platform conventions and apply one visual language throughout (Nielsen, 1994).

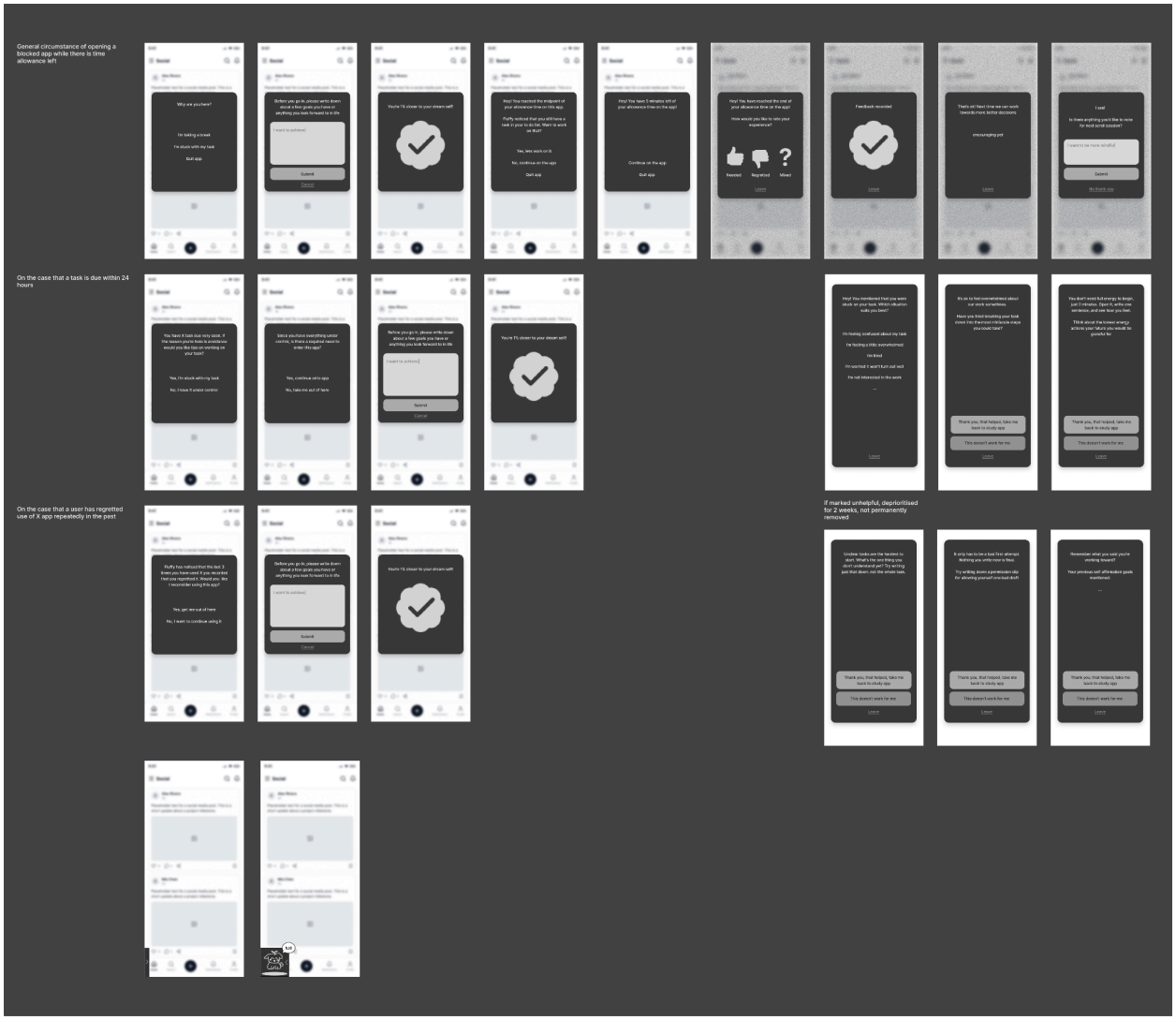
Internally, every intervention, whatever triggered it, appears as the same light-blue card over a dimmed version of the blocked app, anchored by the pet companion in a fixed position. A user who has seen one diagnostic screen recognises the shape of the next, even when the copy changes. "What's going on right now?", "You have [task] due tonight," and "You've marked your last 3 sessions as regretted" are different situations delivered through one template, not three patterns to learn. Button colour is equally fixed where the lighter green always marks the lower-friction, continue-leaning option, and the darker green always marks the action that exits the blocked app. This holds across the whole sequence, including the allowance-warning and end-of-session rating screens, so users can anticipate what each colour means partway through.

The Tips screens extend this principle. Five stuck-states (overwhelmed, tired, confused, worried, not interested) share one layout of heading, short reassurance, action buttons and consistent navigation bar. This was a deliberate rejection of five bespoke screens, each needing separate learning, which would have made the app feel inconsistent and unpredictable, and weakened the impression that it responds intelligently to context.

Externally, the intervention aligns with conventions users already know from the operating system's screen-time tools and from other apps' notification patterns, rather than inventing an unfamiliar interaction model. The types of buttons and text links and style guide is consistent with existing design systems maintaining users' mental model of navigating through information.

This matters because our interview data warned that repeated, poorly designed prompts get ignored or dismissed with frustration. Consistency does not reduce how often the app intervenes, but it lowers the cognitive cost of each intervention, which determines whether a user engages with the fifth prompt as carefully as the first.

Wire-frames



Wireframes for locked out sections and broken down by context specific flows

UBE Data Collection

Participant 4

SUS Score: 97.5

Background: 4rd year university student with ADHD, majoring in social studies, currently on placement

Onboarding

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	Correctly worked out “messaging and calls” after a moment’s confusion about which messages it meant. Asked why she’d add friends and whether it’s a competition, then skipped. She zoomed past the onboarding and hence when some sections were confusing. Didn’t understand the add-task step and asked whether the app locks for 15 minutes. Skipped through the pet chat quickly (she attributed this to ADHD).	“I was wondering why it was called Cove, and I was like, what is this?” “What if I’m a fairy?” “I don’t understand the link between the tasks and the locked apps, but maybe I’m not supposed to understand that yet.”	The purpose of the app (pet care plus productivity), the role of friends, and the task-to-lock link aren’t established before users are asked to act. The “messaging and calls” wording is read differently by each participant. Suggestion was to make the onboarding mascot do more explaining of the context of the app.	3/5

Setting Up Task

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have an assignment due soon. You write out your tasks and break the biggest one down into small, manageable steps you think you need to get started.	Asked “why should I talk to Fluffy?” and whether Fluffy is an AI. Worked out that subtask times add up to the task time. Noticed “10 coins per subtask” and immediately saw an exploit. Finished the task successfully.	“If I’m broke, I should just make fake subtasks.” On prevention: “I don’t think you can, unless you get AI to verify... people can just say they’ve done stuff.” Later: it’s not really a problem. In the post-session notes, she wanted tasks and subtasks connected to Google Calendar.	The pet chat prompted a question about whether it’s an AI, so expectations of the pet’s role weren’t set.	1/5

Starting a Focus Room

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You want to get into a focused headspace, so you decide to start a study session on your own.	Couldn’t find where to start a session at first and expected a “start focus session” button. Found “private room” after some searching. Said she’d only invite	“Shouldn’t there just be a button that says start focus session?” “I never study with people... I wouldn’t think about blocking websites if I’m studying with them online.”	The user's mental models can be different at times as some will look for the rooms directly in navigation while others expect it to be a more visible primary button. Social rooms don’t fit her habits, and she’d	4/5

You suddenly remember that your friends, Jimmy and Mary wanted to study to so you invite them to your session After a while, Jimmy has to leave but you still want someone present so you join a public room After you send the session you are greeted with a reflection activity	friends if physically together. Found the task to work on without help. Said she'd never join a public room, though she tried the flow, and hit a glitch on the exit-session confirmation. Said she'd never do the reflection.	On public rooms: "I don't talk to strangers... what if they're not working?" "I like that the blocking stays on during the break." On the reflection: "I would never look at these reflections." Suggested a scale from 1–10 or faces instead.	rather study in person. Blocking during breaks is valued. The reflection step is likely to be skipped or rushed.	

Pet Playground

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty Rating (1-5)**
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	She found the dress-up items and bought them easily. Identified shop, outfit and house icons, and wasn't sure about the play icon at first (thought it was a	"It would kind of be convenient if you could just press unlock and then 'buy for 30 coins' came up." "What about the people that have ADHD and get sick of having the dog?"	Customisation works. Locked items could offer a direct purchase prompt. Pet choice feels locked-in, and the play section needs a clearer purpose and cost model. Her emotional reaction to the play concept shows the pet	2/5

	ball). Noticed that some items were locked, and found them easy to tell apart. Asked whether play costs coins. Asked to be able to change or add pets.	Suggested multiple pets (“a little family”) and said it might keep her using the app. “That makes me feel a bit depressed... relying on me to work so we can run around.” (on the play concept)	mechanic can feel like guilt-driven care for some users.	

Locked Out Interaction

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty Rating (1-5)**
You set your time limits during onboarding earlier. Right now you have some scroll-time allowance left, and you open Instagram to take a short break and catch up with friends. You have an assignment due in 12 hours. You open Instagram anyways, just to see what is new. You have marked the last	Asked why the check-in appeared, wondering whether she was in a focus session or over her limit. Proposed her own purpose options. Didn’t notice the tired pet until told. Found remaining time and understood the end-of-allowance screen. Chose “I’m stuck / overwhelmed” and found the tips easy to reach. Didn’t treat the regret intervention as a	“You know what would be good if you could customise your options... I’m messaging someone, I’m finding something to show someone.” “It would just get really fast at typing.” “I never noticed that [the tired pet].” Wanted the pet bigger and central, and the energy bar always at the top: “I feel like I would be more mindful of my use if I saw it all the time.” On the 0% lock: “What if my mum texted me... or my future husband messaged me with his proposal plan and I couldn’t open it?” On the language: “I think this is good.” On	The check-in doesn’t fit users who open Instagram for messaging. Written friction would be bypassed. Energy bar and pet state are too easy to miss. The hard lock raises an emergency-access concern but messaging is still possible but it was forgotten. Tips and tone are well received.	3/5

3 sessions as regretted on Instagram but you open Instagram anyways.	privacy issue. Said she'd skip the goals/values reflection and type fast. Asked whether she could unlock at 0%.	the tips: "I think you need more than these three."		

Participant 5

SUS Score: 87.5

Background: Highschool student who scrolls to pass time when bored but is motivated to achieve high scores and do well in her studies.

Onboarding

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	Read "messaging and calls" as notifications until the facilitator corrected her. Added friends and a task without trouble. Didn't notice the coin system.	"I didn't know there was a coin system." "If you make it more clear [during sign-up] that there's a coin system, I think that's also more interesting for people." Said the rest of the sign-up was fine.	Mostly smooth, but two hidden systems (coins, notification permissions) surfaced only when pointed out. The mascot is an existing channel for explaining coins and expectations.	2/5

Setting Up Task

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have an assignment due soon. You write out your tasks and break the biggest one down into small, manageable steps you think you need to get started.	Found the add-task flow and understood subtasks as steps towards the main task.	“It’s the tasks I have to do for that main task.” Suggested linking big tasks to the calendar and letting users add subtasks manually.	The core model is clear. Both asked for calendar integration (Bec in the later part of the session, Participant 4 as an end-of-session suggestion).	2/5

Starting a Focus Room

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You want to get into a focused headspace, so you decide to start a study session on your own. You suddenly remember that your friends, Jimmy and Mary wanted to study to so you invite	Found a private room and invited friends without difficulty, and was confident the invite went through. Expected to tap the whole to-do row to choose a task instead of a check box. Found the public room flow natural. Didn’t realise she	“If it’s group work, or there’s something I’m not confident in and want to ask them.” Would use the focus room for any task, since it blocks apps. After learning about the message limit: “I wanna see how much work I got done with my friends... ‘where are you up to?’” On the reflection: “I’d probably not [reflect]... one question with notes” and keep it optional.	Session flow and invites are clear. The no-chat rule wasn’t expected, but the progress-update use case made sense once explained. She accepts reflection only if it’s optional and lightweight. Tap targets differ from her expectations.	1/5

them to your session After a while, Jimmy has to leave but you still want someone present so you join a public room After you send the session you are greeted with a reflection activity	couldn't chat freely in a session.			

Pet Playground

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	Customised the pet without difficulty. Identified the shop, hanger (outfit) and painting (background). Didn't know what the last icon (play) did. Understood locked items from other apps.	"Yeah, I don't know what the last one is." Said everything made sense otherwise, and that locked items weren't confusing.	Customisation is easy and shop conventions are familiar. The unbuilt play icon is a label gap that will need an icon or text label when built.	1/5

Locked Out Interaction

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You set your time limits during onboarding earlier. Right now you have some scroll-time allowance left, and you open Instagram to take a short break and catch up with friends. You have an assignment due in 12 hours. You open Instagram anyways, just to see what is new. You have marked the last 3 sessions as regretted on Instagram but you open Instagram anyways.	She understood the check-in and liked it. Said she'd have noticed the tired pet, and suggested moving it to the top. Liked messages at different intervals. Chose the stuck-state option and found it easy to reach. Said the regret screen felt fine.	"It keeps sending me a reminder that I need to get back [to my task]. If it wasn't there, I'd probably just continue scrolling." "[A top bar would let me] continue scrolling until it eventually dies... but if it keeps popping up, it might encourage me to get back." Later: "Can't you have both?... Maybe not too many reminders." On the due-soon message: "I do feel attacked, but I think that's good." On the regret screen: "That's the point of the app."	The staged intervention design works as intended. Tone is acceptable, and she sees pressure as helpful. Bar visibility is a trade-off between awareness and anchoring to the limit. The pet cue is easier to notice than for Bec, but still a candidate for relocation.	2/5

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AI (Claude) was used to research literature on effective methods of doomscroll prevention and habit breaking. Synthesis and interpretations are mine.

Shevy Gupta

Task scenario

Persona: Aisha Chen, “The Self-Aware Strategist”

Age: 20

Occupation: Second year commerce student

Quote:

“I always know I’m avoiding something. I need something that doesn't let me argue my way out of it."

Background

Aisha understands her habits well. She has tried Screen Time limits, app blockers and Forest, and each failed. Dismissing the limit takes one tap so she does it without thinking. She holds the off switch, so the moment she wants to scroll she can override the tool meant to stop her. She wants to complete her tasks without feeling judged and to make the moment of giving in a conscious choice rather than a reflex.

Goals

- Wants to build a study habit that doesn't rely on her mood or motivation that day
- Wants external structure without feeling punished or infantilised by it
- Wants proof that avoiding distraction is actually doing something, not just abstaining

Frustrations

- "I set a 20-minute Instagram limit and just tap 'ignore limit' immediately"
- Forest trees dying feels low-stakes - she's uninstalled it twice with no real consequence
- Dislikes tools that nag or shame for giving in or setup which feels too admin

Scenario: Setting up Cove and first few days of using it

It is Sunday evening and Aisha has a Marketing essay due Thursday. She hasn't started. A friend sends her a link to Cove and she downloads it expecting another time limit app she will end up ignoring.

Aisha opens a cover screen with the Cove logo and a cardboard box tied with a ribbon. She taps sign up and on the next screen the box shakes slightly, as if something is inside. It makes her curious rather than impatient. On Choose your companion she swipes between a dog, a cat and a bunny. She picks the dog and names it Fluffy. Fluffy then asks what to call her, which makes the setup feel like a conversation rather than a form.

Fluffy explains that the apps she picks will be blocked once the daily limit is up. Aisha switches on Instagram and YouTube. Messages and calls are marked as always allowed, which reassures her that Cove won't cut her off from her uni group chats.

She sets a daily limit of 2 hours. The line beneath the slider says Fluffy's energy drains while she is scrolling in a locked app and that once it hits zero the app locks until tomorrow. The app's intentions are clear and Aisha onboards smoothly.

Aisha adds her first task: the Marketing essay, category Study, due Thursday, with one small subtask, "Write one rough essay point". On the Pet screen she sees Fluffy in a small room, with her offline time today, an energy bar and a coin balance of zero. The shop sells a ribbon but coins come from finishing tasks, so she opens a private Focus Room, picks her subtask and starts writing.

Twenty minutes later she is stuck and opens Instagram out of habit. Fluffy appears over the dimmed feed and asks what is going on right now. Aisha looks for the dismiss button as she always does, and finds three options: "I'm taking a break", "I'm stuck with my task" and "Quit app". She picks "I'm stuck". Cove asks what is making the start hard, she chooses "I'm worried about quality", and the tip tells her to draft a placeholder, not a verdict. She leaves the app and writes a rough paragraph.

On Monday afternoon she opens Instagram for a genuine break. She taps “I’m taking a break”, and Cove asks her to write down a goal before the feed loads. She types a line about finishing the essay and goes in. The pause is small, but it is the first time a tool has made her say why she’s there. Halfway through her allowance a message notes her unfinished task and asks if she’d like to work on it. She taps “No, continue on the app”, and nothing punishes her for it. With five minutes left, Fluffy appears tired in the corner of the feed.

When the 2 hours are up, Cove asks her to rate the experience as needed, regretted or mixed. She chooses “regretted” and is told that not every scroll needs to be productive. Instagram then locks until tomorrow. Aisha tries to open it again and finds nothing to tap through. She can still make choices during the day, but she can’t talk her way past the end of it.

That evening she finishes her subtask in a Focus Room. When she ends the session, Cove asks for a short reflection before awarding coins. She writes two sentences and spends the coins on a pink ribbon for Fluffy. Fluffy’s energy hasn’t refilled, and Aisha realises it isn’t meant to. Coins reward the work. They aren’t a way around the limit. Aisha finds herself benefitting from redirection and feedback, and with no way to override her daily limit, she is able to make more conscious choices.

Design precedent

Finch (Onboarding) and Tamagotchi (Pet/home screen)

Both precedents rest on the same idea; people look after something they feel responsible for. Cove borrows Finch’s way of starting that relationship and Tamagotchi’s way of keeping it visible.

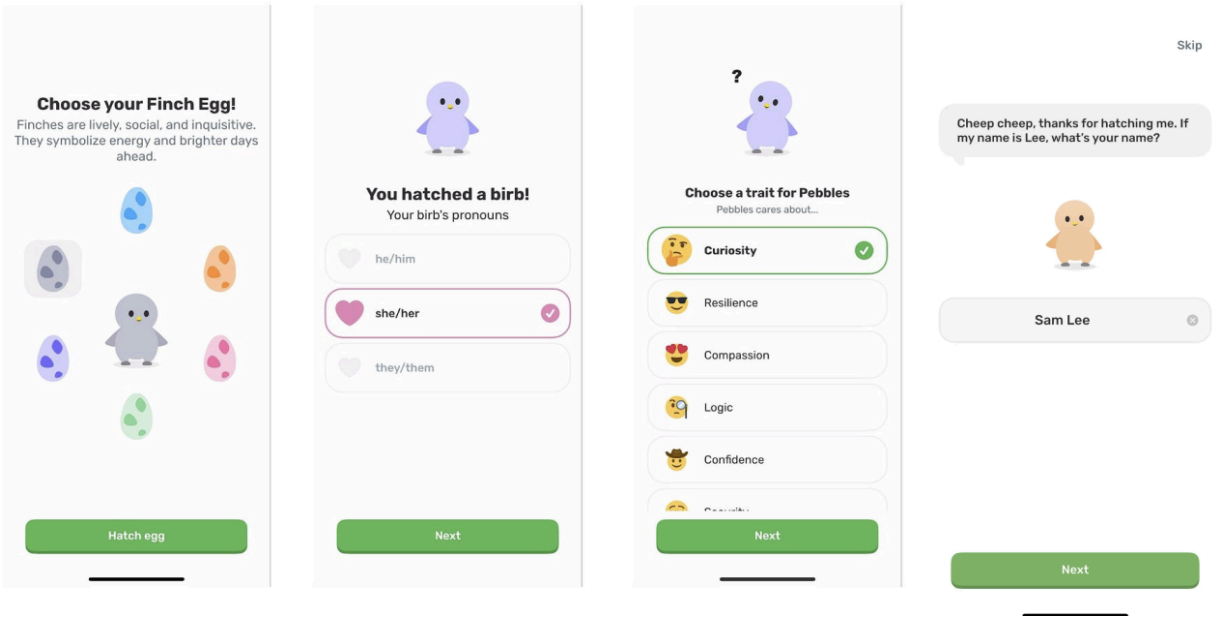
Finch is a self care app where users adopt a virtual pet and care for it by completing real life goals, which gives the pet energy to grow and explore (Finch Care Public Benefit Corporation, 2026). Users choose and name their pet first, and a conversational dialogue with the pet ‘birb’ guides setup including asking for the user’s name, so onboarding feels like meeting someone rather than filling in a form.

Its weakness however is length. The flow runs for 30+ steps and a design critique argues that this overwhelms users who want to start quickly and violates progressive disclosure, which suggests revealing complex features gradually (Hailu, 2025).

Cove keeps the relationship first opening; a gift box, a choice of a companion and a pet that asks for the user's name in its speech bubble. Each step is conversational and the flow is cut down to four steps, with a progress indicator and a skip option on the add friends and task steps.

Tamagotchi was released by Bandai in 1996 and it showed that a simple virtual creature could change real behaviour. This is known as the “Tamagotchi effect”, where people form genuine emotional attachment to a non sentient machine/artificial entity (Zoglmann, 2026). The pet’s current need is visible on screen so users can read its state and feel responsible for it. The artificial sense of responsibility and attachment is enough to drive consistent habit change, even without any tangible reward or punishment.

Cove’s Pet screen applies the same principle. Fluffy is in a room that the user can furnish and dress, where it greets the user and shows an energy bar and coin balance. When the time allowance is nearly used, the same pet appears drained inside the blocked app, so its condition communicates remaining time (visibility). Cove also inverts Tamagotchi’s direction. The original pulled attention toward the device where users devoted more time to checking in on their pet, however Cove uses the same care instinct to support time away from distracting apps. The risk is that attachment becomes compulsive checking so rewards (coins for shop) are earned through focus sessions with a short reflection rather than just through opening the app. Pet interactions are also minimal in Cove for this purpose.



Finch onboarding process - conversational journey with the pet 'birb'

Tamagotchi pet screen - the 'tamagotchi effect' is an emotional attachment people form with their virtual pet

Usability rational

Visibility

Norman (2013) argues that the relevant parts of an interface must be visible, and that they must convey the right message about what the system is doing and what the user can do. Visibility can fail in two ways; the important thing is hidden or it is visible but suggests the wrong meaning. This principle explains the main changes between my preliminary and final designs of the pet screen and onboarding.

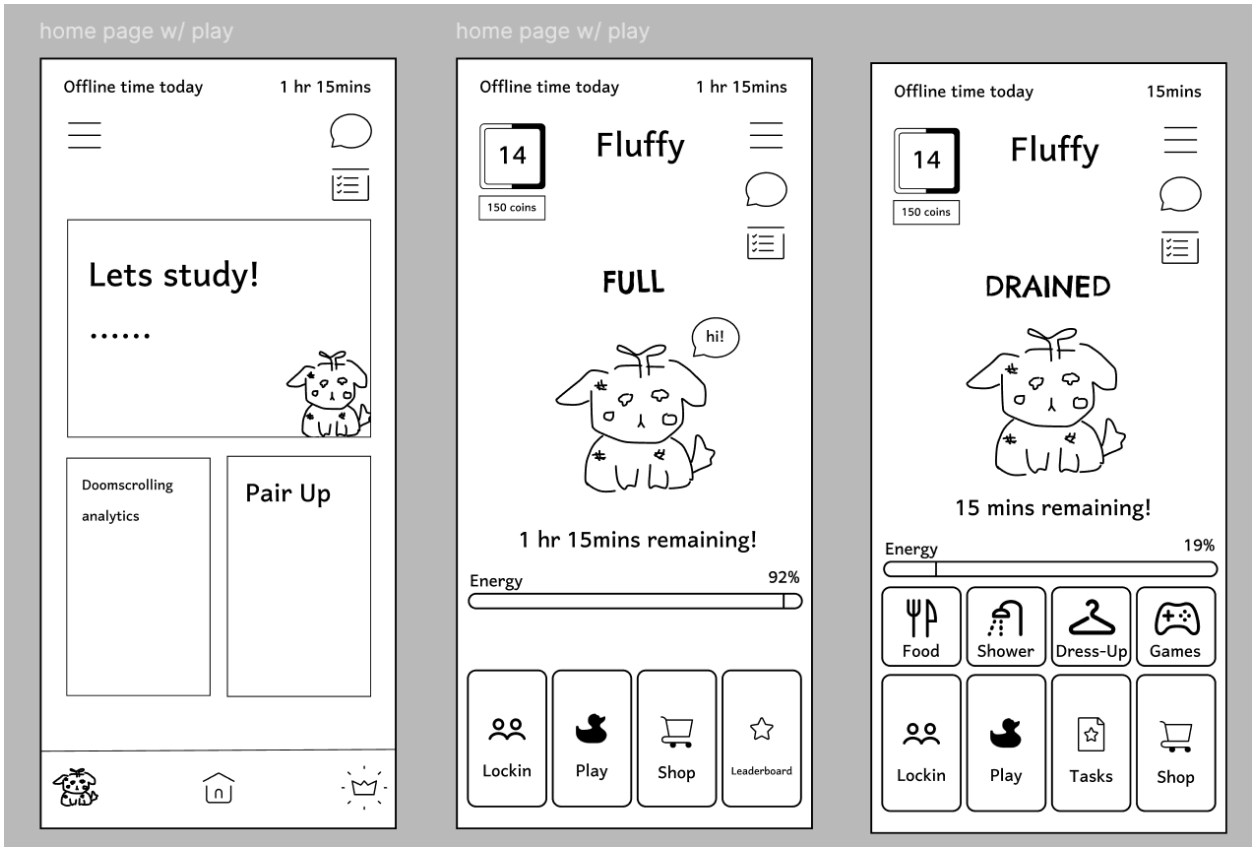
My original Pet screen doubled as the home screen. It showed the pet with a ‘full’ or ‘drained’ label, time remaining, energy, coins and a row of four main actions (Lock in, Play, Shop and Tasks). Play opened a further set of pet-care options: Food, Shower, Dress Up and Games. This created two visibility problems. First, the pet was the focal point of the screen where users would also look for what to do next, so tasks, which matter most to Cove’s purpose, were one tile among four rather than the clearest thing on the page. Second, the care options conveyed the wrong message. Food and Shower belong to Tamagotchi style care where tending the pet restores it, but in Cove energy only drains and cannot be restored, and coins from tasks only unlock customisation. A user could expect Food to refill Fluffy and find that it doesn’t.

The final designs split these screens in two. Home shows what the user needs to act on; a pet card with energy and coins, upcoming tasks with their coin values, and friends to join. The Pet page holds the room and four actions (Shop, Dress Up, Decorate and Play). Actions implying energy could be restored were removed. This keeps task information in front of the user and pet interaction minimal, which also reduces the Tamagotchi risk of compulsive checking.

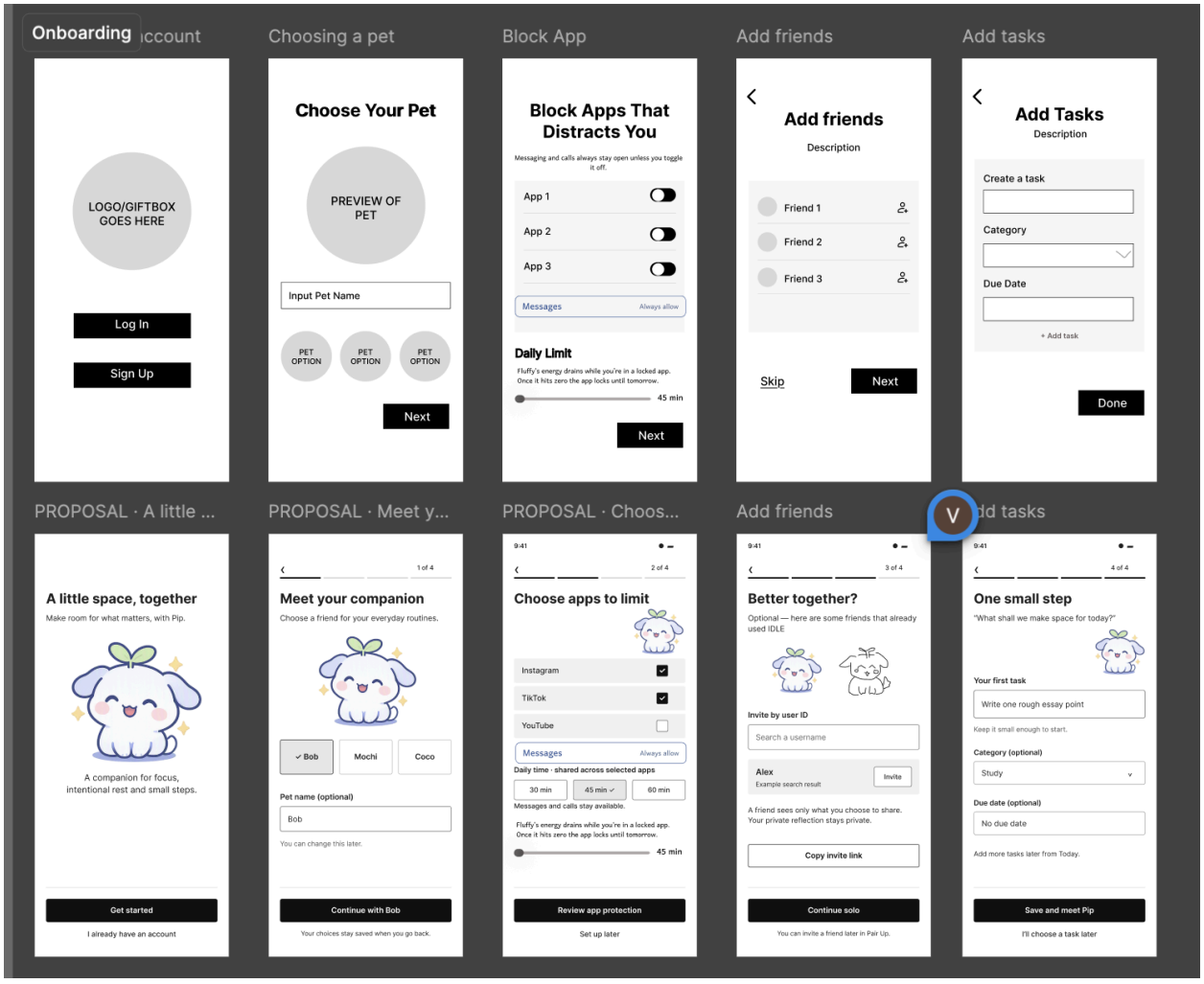
For onboarding my preliminary screens were form like, and the pet appeared only when choosing a companion. The rule that energy drains and apps lock until tomorrow sat in small text beneath a slider. In the final designs the pet appears in every step with a speech bubble explaining that step in plain language, so guidance is where the decision is made. Apps to be blocked are highlighted and progress segments show how much setup remains. Keeping the pet in the same position at each step makes guidance predictable.

Separating the screens has a cost. In group testing, one participant did not discover coins and customisation until after completing their first task, so these are still not visible enough. Home shows coins but not what they are for, so a short prompt after the first completed task could help.

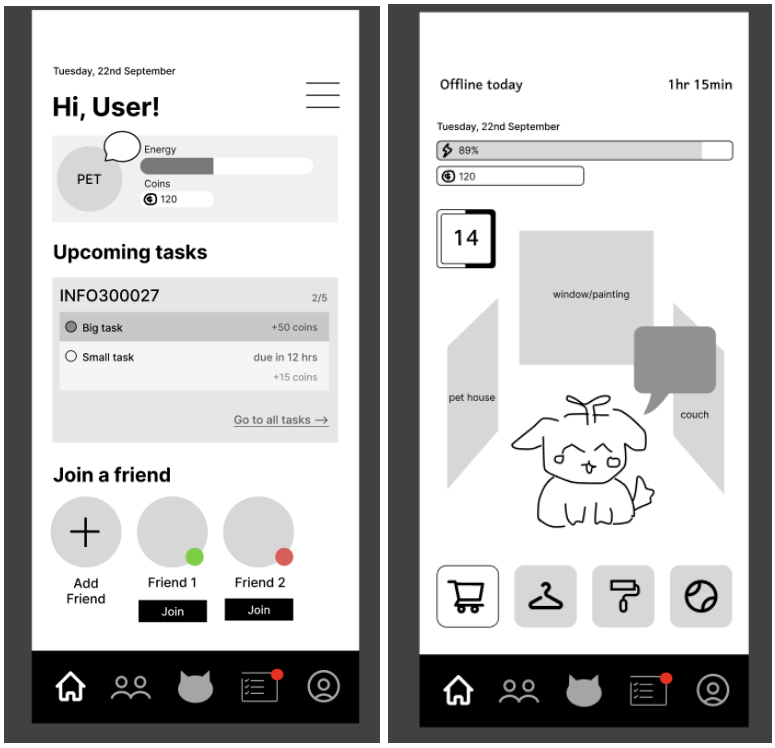
Wire-frames



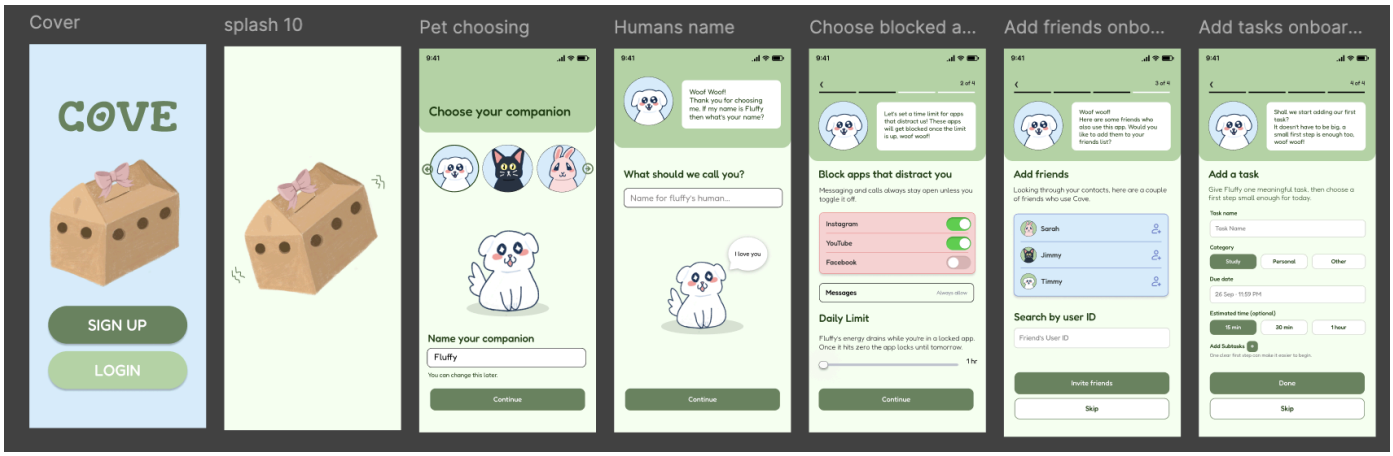
Preliminary designs - 2 homepage versions (one with pet and one without)



Preliminary designs - onboarding versions (second row designed with visibility in mind)



Final designs - homepage + pet page seperate



Final designs - onboarding flow

UBE Data Collection

Participant 2

SUS Score: 92.5

Background: A second year university student studying law, who sometimes uses productivity tools to try and lock in.

Onboarding

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	Easily followed the flow and understood the process. Pet dialogue helped navigate through screens more cleanly + intuitively	“It was pretty easy to follow. Looks cute. The pet dialogue reminds me of some other apps I’ve used. I would download this.” “Cool that you can add friends and motivate them”	She linked the pet dialogue to other apps she used so it draws on a mental model	1 - Very Easy

Setting Up Task

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)

You have an assignment due soon. You write out your tasks and break the biggest one down into small, manageable steps you think you need to get started.	Step breakdown seemed to feel natural to her (followed the flow seamlessly)	“I can see what I need to finish first. It's better to have a task breakdown than not to cos sometimes you do need it.” “I like the reward system, getting coins and customising pet. Its fun to do. If I don’t have rewards im not gonna do tasks so its motivating”	Her comments suggests subtasks help when a task is large but shouldn’t be forced so it should be optional	1 - Very Easy

Starting a Focus Room

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty Rating (1-5)**
You want to get into a focused headspace, so you decide to start a study session on your own.	Flow was easy to follow however she didn’t like the restrictions placed. Preferred to have scrolling apps unblocked during break, and wanted more than 5 mins for break. Was not keen on reflections either.	“I like that there’s automated messages in focus sessions. Helps to keep focus but allows you to still motivate friends.” “I always scroll during break so I don’t like that. I would just put random letters for the reflection bit cos I don’t want to reflect all the time”	If users can type anything to bypass reflection then it may not show real engagement. We might wanna reconsider length and whether its mandatory, or whether there are checks in place to ensure user is typing meaningfully	3 - Neutral

Locked Out Interaction

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You set your time limits during onboarding earlier. Right now you have some scroll-time allowance left, and you open Instagram to take a short break and catch up with friends.	Flow was okay but its noted that she felt if pop ups were regular, she might get annoyed (every time opening an app, she wouldn't want there to be a popup) She liked having pet and energy in corner to tell her screen limit	"Why are you here pop up is good. It reminds me that I have tasks to complete. If this was regular tho I would probably get annoyed." "pet icon is nice or somewhere to say where I have time left"	Frequency matters as much as content, so the blocked app prompts might need to appear less often for users who open it regularly. She also valued the pet and energy in the corner as a way to see time left	3 - neutral

Participant 9

SUS Score: 87.5

Background: A third year university student studying occupational therapy, who struggles with doomscrolling and starting assignments on time.

Onboarding

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)

A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	The flow was straightforward, however she skipped through some of the onboarding dialogue. She was also curious about what coins represented and how sub tasks worked	“I’m finding this not very confusing to use, its telling me what to do” “It’s aesthetic and the pet is cute” “I’m curious what the coins represent hmm”	If users skim the speech bubbles, key rules (what coins are for, how energy works, etc.) need to appear somewhere they can’t miss	2 - Easy

Setting Up Task

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty Rating (1-5)**
You have an assignment due soon. You write out your tasks and break the biggest one down into small, manageable steps you think you need to get started.	She was able complete the flow naturally and navigate to her tasks from homepage too	“Oh ok breaking it down so sub tasks” “Pretty easy to see tasks from the homepage”	Finding her tasks from the homepage was easy, which supports putting upcoming tasks on Home. No usability issues were observed here.	1 - Very Easy

Starting a Focus Room

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty**

				Rating (1-5)
You want to get into a focused headspace, so you decide to start a study session on your own.	Figured out the flow easily and compared it to zoom. Struggled with reflections since she wasn't used to doing one on other apps. She also wanted to earn coins without reflections	"I think its in focus room, then start private room, oooh this is kinda like zoom" "Send a message that's nice" "Awhh I want coins without reflection" "I'm bit unfamiliar with the reflections however overall it's pretty easy to use"	The room borrows a mental model she already had (Zoom). The reflection was the main friction. She was unfamiliar with it and wanted coins without it.	2 - Easy

Pet Playground

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have completed a focus session and earned coins. Explore the Pet Playground to see how you can use your	She was able to easily identify what different buttons were for in the pet playground. However, was slightly confused with what	"Ok this looks like the customise place" "Ball is probably playing with the pet, and paint is maybe the background"	She correctly guessed the purpose of the buttons from their icons so the Pet page actions are well signified.	1 - Very Easy

rewards to interact with or customise your pet.	the energy was since she did not read that onboarding step clearly	“I guess I have to study and complete more tasks to earn coins”	Energy’s meaning should be visible on the Pet page itself not only in onboarding.	

Locked Out Interaction

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty Rating (1-5)**
You set your time limits during onboarding earlier. Right now you have some scroll-time allowance left, and you open Instagram to take a short break and catch up with friends.	She found setting intentions before using a blocked app meaningful. Redirections were nice however she noted that for some people these techniques might work, while others might get rage baited and feel inclined to delete the app.	“I think it would be pretty great for someone trying to limit their screen time” “Little reminders is really good and can help me get back on track” “For some people it might be a little rage inducing to keep having to write intentions down when opening a blocked app”	The interventions seem helpful for users who want help, but frequency needs tuning so they don’t push people away	1 - Very Easy

Main Findings

Participant 2 found Cove easy to use (SUS 92.5) and responded positively to the pet dialogue, task breakdown, coin rewards and the collaborative focus room. She followed onboarding and task setup without difficulty and appreciated seeing her remaining time through the pet and energy in the corner of a blocked app. Her main concerns were autonomy however. She disliked the short break limit and wanted scrolling apps unblocked during breaks. She also mentioned how she would enter random text to skip reflections and felt repeated “why are you here” prompts would become annoying in the long run. This suggests the app works best when its restrictions are kept light and optional.

Participant 9 found Cove easy to use overall (SUS 87.5) and completed every task with little difficulty. She worked out the focus room by comparing it to Zoom, and easily identified the Pet Playground actions from respective icons. There were two issues however. She skipped parts of the onboarding dialogue which left her unsure what coins and energy represented. She also wanted coins without providing a reflection.

Ultimately, she valued the intention setting prompt in the locked out flow but also felt that repeated prompts could frustrate some users enough to delete the app.

References & Statement of AI use

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Statement of AI use

AI has been used to ideate potential personas, however, scenarios and research has been conducted on my own.

Violani Julia

Task scenario

The Screen Time Monitor — "*I can see how much time I spend scrolling compared to studying, but I still struggle to change the balance."*

Name: Lyla

Age: 20

Occupation: A full-time university student

Primary Intention: To turn awareness of her screen time and study habits into more intentional digital behaviour and healthier study routines.

Background

Lyla is a university student who regularly checks her phone's Screen Time statistics to monitor how much time she spends on different applications. She also uses a study duration tracker to record the hours she spends studying. By comparing these two records, she tries to understand whether she is spending her time effectively. However, she often finds a significant gap between her screen time and her recorded study hours. Although she recognises this imbalance, she struggles to consistently change her habits, particularly when assignments feel overwhelming or difficult to begin.

Goals and Motivations

Lyla wants to develop a healthier balance between scrolling time and studying. She is motivated by tracking her habits and seeing measurable progress, using her phone statistics and study records to evaluate how she spends her day. She wants to reduce unnecessary scrolling, increase meaningful study time and understand what causes the gap between the two. Rather than completely eliminating social media, she hopes to develop more sustainable habits that help her stay accountable without feeling pressured to be productive every moment.

Pain Points

- **Awareness without action:** Lyla can see the gap between her screen time and study hours, but the numbers alone do not help her change her behaviour.
- **An imbalance between scrolling and studying:** She sometimes spends significantly more time on her phone than she intends, leaving less time for academic work.
- **Avoidance of difficult tasks:** When academic work feels overwhelming or she is unsure where to begin, she often turns to social media for temporary relief.
- **Limited understanding of her habits:** Her trackers show how much time she spends on each activity, but not always why she gets distracted or what triggers her scrolling.
- **Frustration with inconsistent progress:** Even when she sets study goals and tracks her hours, she struggles to maintain the habit she wants.

Task Scenario

Lyla is preparing to work on an upcoming university assignment that is due in four days. One evening, she checks her phone's Screen Time statistics and discovers that she has spent nine hours on her phone that day. She then opens her study time which shows that she has only recorded three hours of studying. Seeing the difference frustrates her because she knows she could have used some of that time to make progress on her assignment.

Determined to improve, Lyla plans to spend more time studying the following day while keeping her scrolling habits in check. She begins working on her assignment but occasionally opens her TikTok when she feels the need for a break. Although she returns to studying afterwards, she finds herself repeating this cycle throughout the day. Wanting to make up for the time she spends scrolling, she tries to fit in more study hours, while still feeling tempted to check her phone whenever her work becomes difficult or boring. By the end of the day, she is unsure whether she has achieved the balance she intended.

Lyla regularly compares her Screen Time statistics with the hours recorded on her study timer to evaluate how she spends her time. However, focusing on balancing the numbers does not necessarily help her understand when scrolling is a genuine break and when it becomes a way to avoid work. She needs

support in recognising these patterns, managing her breaks more intentionally and returning to her tasks without feeling that she must compensate for every minute spent on her phone.

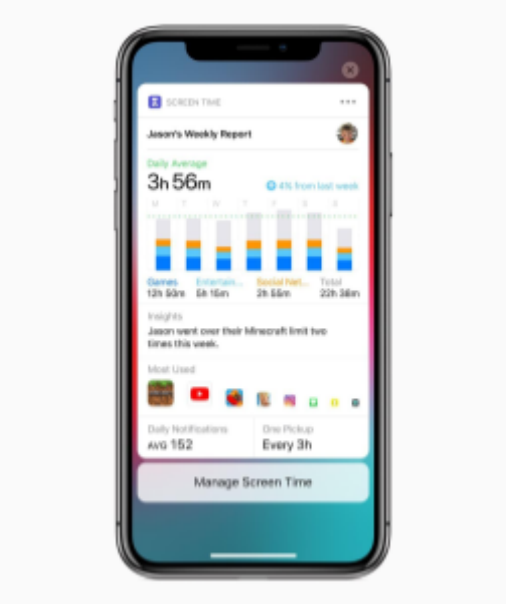
She discovers Cove, an app designed to help students manage distractions and overcome procrastination. She sets up a blocker for her distracting applications, adds her assignment to the task list, and the app helps her to identify the first small step to start studying. When she tries to open TikTok during a focus session, Cove interrupts her and prompts her to reconsider why she is leaving her task. Through the intervention and the tips provided by Cove, Lyla begins to recognise that she sometimes turns to scrolling when she feels overwhelmed or stuck on difficult work. Instead of continuing to scroll, she decides to return to her assignment and try a different approach.

Over time, Lyla uses Cove alongside her existing screen-time and study trackers. She begins exploring her scrolling patterns through Cove's analytics and reflections, discovering how her emotions and study experiences influence her habits. She becomes more aware of when she scrolls to relax and when she uses it to avoid uncomfortable feelings associated with academic work. By reflection on these patterns over time, Lyla can better understand her behaviour and identify strategies that help her refocus. Rather than focusing solely on balancing her screen time and study hours, she begins making more intentional decisions about when to take breaks, when to return to work and how to manage moments when she feels overwhelmed.

Design precedent

Cove's Account page draws inspiration from Apple Screen Time analytics and Spotify Wrapped, combining practical self-monitoring and personal reflection with engaging data storytelling .

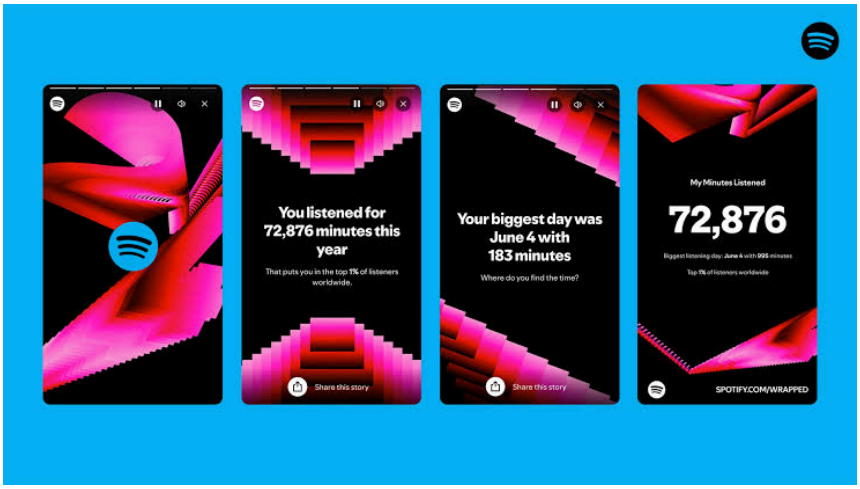
Apple Screen Time: Monitoring and Comparing behaviour



Apple Screen Time Analytics IOS 12, The Sweet Setup

Apple Screen Time provides daily and weekly reports of app usage, allowing users to identify where their time is spent. This influenced Cove's use of usage metrics and comparisons across time in blocked apps and productivity in Cove. Recent research on shared personal informatics continues to draw on a five-stage model of self-tracking, encompassing preparation, data collection, integration, reflection and action (Xue et al., 2025). This suggests that presenting statistics is only one part of supporting behavior change. By comparing current screen time with previous periods, Cove aims to help students identify changes in their habits and evaluate their progress. Research on learning analytics dashboards also suggests that carefully designed comparisons with relevant peers can support motivation and academic achievement, although effects depend on how comparison is implemented (Fleur et al., 2023). Rather than comparing users with other students, Cove focuses on comparison with users' own past behaviour. This allows users to recognise personal progress without the pressure of competing against others.

Spotify Wrapped: Personalised data storytelling



Spotify Wrapped 2024, BBC

Spotify Wrapped transforms listening statistics into personalised, visually engaging stories. Rather than presenting data as a conventional dashboard, it encourages users to explore their habits through individual cards, inspired by the story format used on platforms like Instagram. This inspired Cove's Account page, which presents scrolling patterns, app-blocking attempts and progress through individual summary cards, organised into weekly and monthly recaps. The aim is to make behavioural data easier to explore and more engaging to revisit. Unlike Spotify Wrapped's celebratory focus, Cove applies this approach to digital well-being to encourage users to recognise patterns and acknowledge progress without feeling judged.

Reflection Recap: Learning from past experiences

Cove's *Reflection Recap* complements analytics by allowing students to revisit previous reflections about why they scrolled, how they felt and what they intended to do next. This gives users an opportunity to recall earlier experiences, learning recurring triggers and compare their previous intentions with their current behaviour. Leyland et al. (2024), revealed that reviewing earlier written reflections helped students identify personal change and developments in their understanding and self-regulation. Cove applies this principle by making previous entries accessible, helping students learn from their experiences instead of treating every scrolling episode as an isolated event.

Together, these precedents inform two complementary functions in the account page: analytics helps students recognise *what* is happening in their digital habits, while Reflection Recap helps them consider *why* it happens and how they might respond differently. Both features aim to support self-awareness and more intentional behaviour over time.

Usability rational

Recognition Rather Than Recall is a usability heuristics that emphasises making relevant information and options visible to users to reduce the need to retrieve information from memory (Budiu, 2024). Nielsen's usability heuristics remain a widely used framework for evaluating interface design, and recognition rather than recall is included among the ten principles discussed in research on human-computer interaction (Nielsen, 1997; Paton et al., 2021). This principle informs the design of Cove's Account page, which combines usage analytics with reflection recaps to help students review their digital habits and revisit previous experiences.

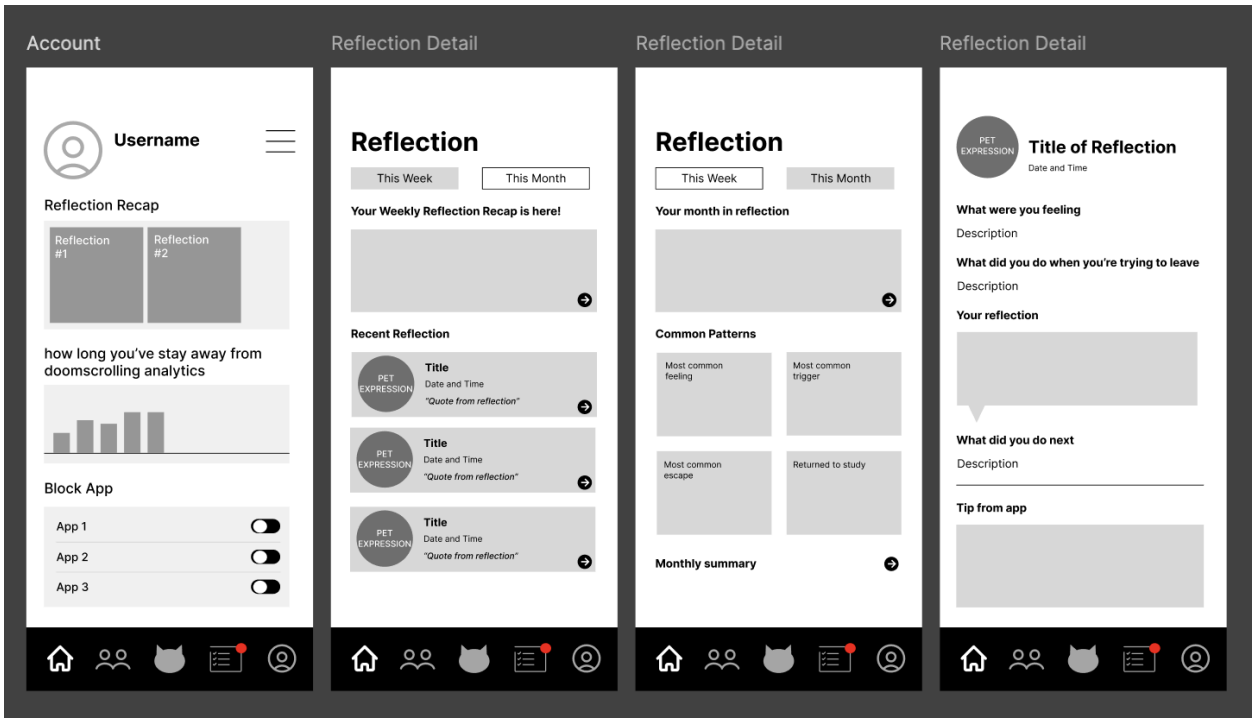
The Reflection Recap presents relevant information in a structured format: what did the user felt, what did they try to do, their reflection in sentences, what did they do next. This approach is supported by Moore et al. (2022) where they identified the analysis of personal data as an important part of personal informatics, highlighting the need for tools that support users in exploring and interpreting their own information. In Cove, side-by-side screen time usage and productivity recorded duration analytics allow students to identify patterns in their screen time and focus activities, while the reflection recap provides an opportunity to revisit previous experiences and consider how their behaviour relates to their intentions.

These features aim to make personal information easier to access and interpret, helping students connect their past experiences with future decisions. However, presenting data alone does not guarantee meaningful reflection. Eikey et al. (2021) caution that engaging with personal data can sometimes lead users to become caught in negative thought patterns rather than gaining constructive insights.

Therefore, Cove's account page should present analytics and reflection recaps in a supportive, non-judgemental manner, avoiding an excessive emphasis on failure or unproductive comparisons. By making past activity visible while encouraging constructive reflection, the design aims to help students

better understand their digital habits without placing the burden of remembering and evaluating every experience entirely on them.

Wire-frames



Initial Ideation of the Account Page

UBE Data Collection

Participant 3

SUS Score: 82.5/100

Background: A university student who prefers studying independently and uses productivity tools to manage distractions and stay on-track. They value external restrictions such as app reminders to help them return to studying when tempted to doomscroll. While they prefer practical tools that directly support their study goals.

Onboarding

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	Participant navigated task creation, including adding task details and subtasks, but experienced some confusion during the onboarding walkthrough.	*"The onboarding is pretty straightforward, classic. But, the task creation section looks a bit complicated, and I didn't even see the part where you can add a subtask."*	Onboarding and task creation could be simplified to reduce confusion, such as removing the adding subtask feature in the onboarding step.	1 = very easy
	Participant followed the setup flow but did not discover the coins and pet customisation features until after completing their first task.	*"It would be good to introduce the pet together with how to get coins to customise your pet in onboarding."*	Introducing the reward system earlier could help users understand how to earn and use coins, making these features easier to discover and potentially increasing engagement within the app.	

Setting Up Task

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)

You have an upcoming assignment and want to break it down into manageable steps so you can stay focused and avoid procrastinating.	Participant successfully added a task, filled in task details, added subtasks and completed the task.	*"I would only create subtasks if the task was complex enough."*	Some users may prefer flexibility in how they organise their work, therefore labeling subtasks as optional will allow users to not feel obligated breaking their task down.	1 = very easy

Starting a Focus Room

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty Rating (1-5)**
Now that you have created a task list for your assignment, start a focus session to avoid distractions. Choose the setup that best suits your study preference.	Participant chose "start privately" for solo study and appreciated being able to select a task and invite others. They found the invitation interface intuitive.	*"I'd usually prefer to study alone so the private focus room would be the best option for me to study."*	Offering both private study sessions and the option to invite others accommodates independent study preferences while still allowing users to benefit from social accountability.	1 = very easy
	Participant perceived public rooms as anonymous and similar to private rooms with friends, questioning whether other users were genuinely focused.	*"How would I know if other people in the other person in the room were actually focusing?"*	A shared focus room may not provide sufficient social accountability when users cannot tell whether others are genuinely studying. Providing clearer indicators of participation or activity could help users feel more confident in the	

			value of studying alongside others.	

Pet Playground

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have completed a focus session and earned coins. Explore the Pet Playground to see how you can use your rewards to interact with or customise your pet.	Participant explored the pet feature but did not find it personally motivating. The coins and pet customisation features were not discovered until after completing the first task.	*"The pet is cute, but it doesn't really motivate me."*	The pet and coin reward system may not motivate all users equally. Introducing additional or alternative rewards may be needed to appeal to different user preferences.	1 = very easy
	Participant suggested making coins more valuable to encourage engagement.	*"It would be more motivating if I could exchange coins for extra doomscrolling time."*	Make rewards more meaningful by linking coins to incentives that support users' motivations for using Cove.	

Locked Out Interaction

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty

				Rating (1-5)
You are in the middle of a focus session when you feel the urge to leave Cove and open a distracting app. Follow the prompts and see how Cove responds to your attempt to leave the session.	Participant understood the purpose of the reminder and lock-out system and indicated they would likely return to the app after being locked out.	*"I would probably return to the app once it locked me out."*	The intervention may effectively interrupt doomscrolling and encourage users to return to their tasks, suggesting that app restrictions can support task persistence.	2 = easy
	Participant recognise that the intervention could help people who need external enforcement to stay focused.	*"It would be helpful for someone who needs something to keep them focused."*	Intervention may be more valuable for users who rely on external structure to manage distractions, although its effectiveness may vary depending on individual preferences and self-regulation needs.	
	Participant indicated that they would likely skip the reflection prompt because they were not inclined to write responses.	*"I would rather if it asked me a word or one short sentence, but most of the time I think I would skip it."*	Written reflections may create friction for users who prefer not to think and write. Retain the pop-up intervention and offer quicker alternative such as more specific options of reasons to leave the focus session.	

(Additional) Account Page

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have been using Cove for a while to manage distractions and stay focused on your university work. You want to review your past activity to understand how your scrolling habits and productivity have changed over time. Explore your account and see what insights you can gain from your progress.	Participant expressed interest in viewing doomscrolling and productivity analytics side by side, noting that changes in the data could motivate them to become more productive over time.	*"When I see some progression in the analytics it does motivate me to keep going."*	Comparing scrolling and productivity data may help users recognise behavioural changes and motivate them to improve their habits.	1 = very easy

Participant 8

SUS Score: 70/100

Background: A university student who experiences procrastination when academic tasks feel difficult or demanding, sometimes turning to digital distractions instead of addressing the underlying challenge. They value quick access to studying and effective app-blocking interventions but find extensive task planning and mandatory reflection burdensome. They prefer flexible productivity tools that minimise setup effort and provide concise prompts to help them recognise distractions and refocus on their work.

Onboarding

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
A friend tells you about a new productivity app. You've been meaning to cut down on procrastinating, so you decide to check it out and sign up.	Participant misunderstood “daily limit” as daily usage quotas to access blocked apps rather than a period of app blocking during study.	*“I thought it was a daily limit usage.”*	Use clearer, consistent language to explain the app-blocking mechanism during onboarding.	2 = easy

Setting Up Task

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have an upcoming assignment and want to break it down into manageable steps so you can stay focused and avoid procrastinating.	Participant found task creation understandable but felt that entering subjects, assignments, subtasks, dates and categories involved too many steps.	*“I just want a simple task list where I can add a task, instead of feeling like I have to do a whole planning before studying.”*	Simplify task creation and making additional details optional could help users begin studying more quickly without feeling pressured to plan everything in advance.	1 = very easy

Starting a Focus Room

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
Now that you have created a task list for your assignment, start a focus session to avoid distractions. Choose the setup that best suits your study preference.	Participant understood the focus-room concept and preferred starting a study session directly rather than having to plan a task list.	*"I'd rather start studying straight away."*	Requiring users to create a task list before starting a focus session may create unnecessary friction. Allowing users to start a focus session immediately, with the option to add a task accommodates those who prefer to begin studying without prior planning.	1 = very easy
	Participant expected public rooms to show profile pictures or chat but questioned whether visible co-working alone would provide accountability.	*"Just seeing other people there with a timer doesn't necessarily mean they're focusing."*	Public focus rooms may not provide sufficient social accountability when users cannot meaningfully interact with or assess other users' engagement. Clarify the purpose of public rooms and consider adding optional social features or visible participation cues to strengthen accountability.	
	Participant suggested occasional progress check-ins during focus sessions.	*"Maybe check in every 30 minutes or an hour."*	Offer optional, appropriately timed check-ins to support focus without creating additional distractions.	

Pet Playground

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You have completed a focus session and earned coins. Explore the Pet Playground to see how you can use your rewards to interact with or customise your pet.	Participant considered pet care and coin rewards insufficient motivation to complete mandatory tasks.	*"It's straightforward."* *"But I feel like personally I wouldn't be motivated enough to finish my task just to earn coins for my pet at the end."*	The reward system may not provide sufficient motivation on its own to encourage task completion. While the pet and coins add a playful element, they may need to offer more meaningful or personally relevant incentives to encourage sustainable engagement.	1 = very easy

Locked Out Interaction

Task-Scenario	Observed Behaviour	Participant Feedback	Usability Insight	Task Difficulty Rating (1-5)
You are in the middle of a focus session when you feel the urge to leave Cove and open a distracting app. Follow the prompts and see how Cove responds to your	Participant responded positively to the blocked-app intervention, particularly the prompt offering tips by going back to Cove which interrupted	*"Sometimes I'm scrolling because I'm stuck on something difficult."*	Interventions that help users recognise when they are stuck and provide guidance for resuming their work may be more useful than app blocking alone.	2 = easy

attempt to leave the session.	doomscrolling and encouraged them to reconsider why they were avoiding work.			
	Participant would likely skip the reflection step because they did not enjoy writing reflections.	*"I'm skipping them, I don't think I would want to write anything, the pops up are enough to give me a reality check and go back studying."*	Some users are unwilling to engage with a more effortful reflection process. Keeping the intervention concise, allowing users to return to their task immediately without being required to write a reflection.	

(Additional) Account Page

Task-Scenario	**Observed Behaviour**	**Participant Feedback**	**Usability Insight**	**Task Difficulty Rating (1-5)**
You have been using Cove for a while to manage distractions and stay focused on your university work. You want to review your past activity to understand how your scrolling habits and productivity have changed over time. Explore your account and see what insights you can	Participant found side-by-side doomscrolling and productivity analytics helpful for behavioural changes over time.	*"If I see changes such as the bar is lower on the doomscrolling, it would make me feel better."*	Visual comparisons can help users recognise changes in their habits and understand their progress over time.	1 = very easy
	Participant felt that reviewing reflection recaps would not personally motivate	*"I think the reflection recap would be helpful for someone who enjoys journaling, however for me, the analytics*	Reflection recaps may appeal more to users who enjoy writing and self reflection, rather than users who	

gain from your progress.	them, although they recognised their potential value for users who enjoy journaling.	*itself is enough to see my behavioural pattern. I'm not really the type to write."*	prefer data-driven feedback.	

Main Findings

The usability tests with both participants revealed that Cove's core interruptive doomscrolling features were valuable, while aspects of its setup flow, feature discoverability and additional activities could create usability barriers. Both participants responded positively to the app-blocking interventions which shows that Cove potentially helps students return to studying when distracted. However, both indicated that they would likely skip the written reflection prompt because they were not inclined to write about their experiences, suggesting that mandatory reflections could create unnecessary friction in the process of returning to work.

The participants also expressed different preferences regarding task planning and progress tracking. While Participant 3 successfully navigated task creation, they experienced some confusion during onboarding. Participant 8 preferred starting a focus session immediately rather than completing extensive task planning. Both participants found the virtual pet appealing but not particularly motivating, indicating that its reward may not be sufficient to encourage task completion on their own. Regarding the Account page, both participants valued side-by-side analytics comparing doomscrolling and productivity, as they believed changes in the data could motivate behavioural improvements over time.

These findings suggest that Cove should prioritise effective distraction interventions, a simplified setup process and flexible ways for users to engage with its feature. Making task planning and written reflections optional, improving feature discoverability, and presenting clear comparisons of scrolling and productivity could accommodate different user preferences. Rather than relying solely on pet rewards or mandatory reflection, Cove should support users in recognising behavioural patterns and returning to their tasks with minimal friction.

References & Statement of AI use

I acknowledge the use of ChatGPT to support the development of this assessment.

I used it to support research, develop and refine ideas.

Suggested sources were independently reviewed and verified to ensure their credibility and relevance to the topic. All final decisions and interpretations were made independently.

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